

EEE 312 (January 2026)

Digital Signal Processing I Laboratory

Final Project Report

Section: C2 Group: 01

Detecting Alzheimer's Disease from Resting-State EEG: A Validated Digital Signal Processing Pipeline

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1 Abstract

Alzheimer's disease (AD) is the most common cause of dementia, yet the tests that confirm it (PET, MRI and cerebrospinal-fluid assays) are expensive and scarce outside large hospitals. Resting-state electroencephalography (EEG) is cheap, portable and harmless, and AD is known to slow the brain's electrical rhythms. This project designs, implements and validates a complete digital signal processing (DSP) chain that converts a 19-channel, eyes-closed EEG recording into a probability of AD. Using the public OpenNeuro ds004504 dataset (36 AD, 23 frontotemporal dementia (FTD) and 29 cognitively normal (CN) participants), we designed and verified zero-phase FIR and IIR band-pass filters and a 50 Hz notch, an artefact-removal stage (common-average reference, independent component analysis and amplitude rejection), our own Welch power-spectral-density estimator (identical to SciPy to machine precision) and 94 regional spectral and time-domain features. A logistic-regression classifier was evaluated with nested, subject-wise cross-validation, so that every prediction was made by a model that had never seen that person. The main analysis was fixed before any result was seen. It separated AD from CN with an area under the ROC curve (AUC) of 0.811 (95 % CI 0.717–0.896), balanced accuracy 0.76, and a permutation p-value of 0.005. The result is carried by EEG slowing: the seven regional theta/alpha ratios alone reached 0.827. An ablation showed that the average reference is the single most important DSP step (+0.13 AUC). All five robustness criteria declared in advance were met, and the EEG remained strongly predictive after controlling for age and sex in three ways (AUC 0.75–0.79). FTD versus CN was moderately separable (0.728), but AD versus FTD was not (0.578). We also found that participant numbers do not match between ds004504 and its companion dataset ds006036. The system is a low-cost screening aid, not a diagnostic test.

Keywords: EEG, Alzheimer's disease, digital filter design, Welch spectral estimation, independent component analysis, nested cross-validation, robustness analysis

2 Introduction

2.1 Motivation

Dementia affects more than 55 million people worldwide, and Alzheimer's disease accounts for an estimated 60–70 % of cases [1]. AD is a progressive neurodegenerative disorder: memory declines first, followed by language, executive function and, finally, independence. No cure exists, but early detection still matters. The available treatments and support work best when started early, a confirmed diagnosis lets patients and families plan care while the patient can still take part in decisions, and clinical trials need patients identified before irreversible neuronal loss.

The most sensitive diagnostic tools are amyloid or FDG positron-emission tomography (PET), magnetic-resonance imaging (MRI) and cerebrospinal-fluid assays. They are expensive, need specialised infrastructure, and lumbar puncture is invasive. In Bangladesh these tools are concentrated in a few large hospitals. EEG has a completely different profile: an EEG test costs roughly BDT 500–1,500 in government hospitals and BDT 1,500–6,000 in private centres [2], it uses no radiation or contrast agent, it is entirely non-invasive, and EEG machines are already installed in most tertiary hospitals for epilepsy work. If a reliable AD marker can be extracted from a routine EEG automatically, a screening tool could run on equipment that already exists.

2.2 The problem as a signal-processing problem

The EEG change caused by AD is not a waveform that can be spotted by eye. It is a gradual redistribution of power across frequency, known as **EEG slowing**: the alpha rhythm (8–13 Hz), strongest at the back of the head with the eyes closed, weakens and slows, while theta (4–8 Hz) and delta (0.5–4 Hz) power grow [3]. Detecting it is therefore a frequency-domain estimation problem before it is a classification problem. The signal must be band-limited without phase distortion, separated from much larger artefacts (blinks of hundreds of microvolts against brain rhythms of tens), converted into a low-variance spectral estimate with enough frequency resolution to separate theta from alpha, reduced to a small number of interpretable features, and finally classified in a way whose accuracy is an honest estimate for new people. Each of these stages is a DSP design problem with competing requirements, and each one changes the input of the next.

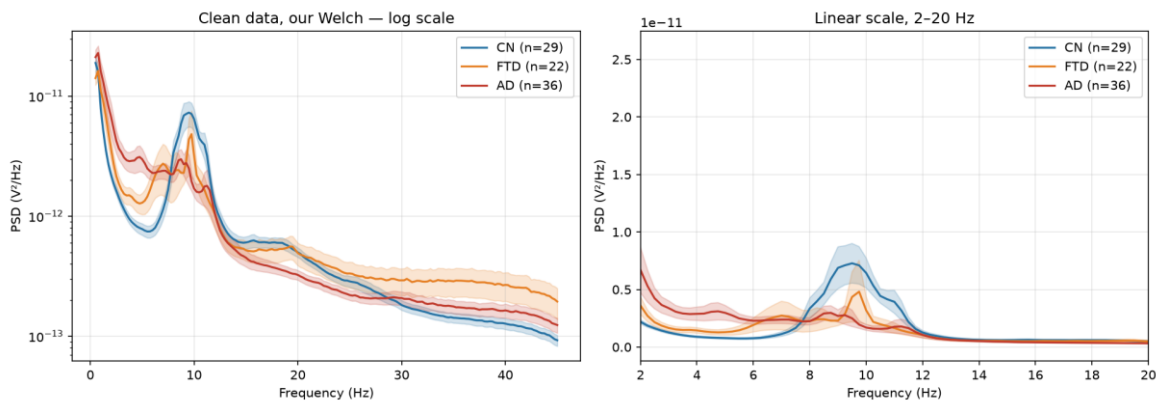


Figure 1. Grand-average spectra of the three groups after our pipeline (log and linear scale): the alpha peak near 10 Hz is lower in AD, and power below 8 Hz is higher

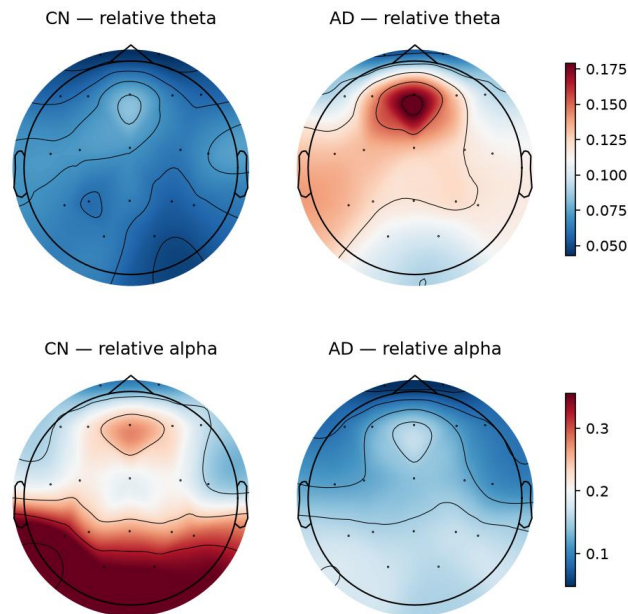


Figure 2. Relative theta and alpha power over the scalp in CN and AD: theta rises, and the posterior alpha maximum collapses, in AD

2.3 Why this is a complex engineering problem

Table 1 maps the project onto the attributes of a complex engineering problem.

Table 1. The project against the attributes of a complex engineering problem

Attribute	How it appears in this project
Depth of knowledge required	Digital filter theory (FIR window design, Butterworth IIR, notch, zero-phase filtering), sampling and resampling, blind source separation (ICA), spectral estimation theory (bias, variance, leakage, resolution), and statistical learning and validation.
Conflicting requirements	Frequency resolution against estimator variance; removing artefacts against keeping brain signal; sharp filter edges against filter length and delay; sensitivity against specificity; model flexibility against overfitting with only 65 people.
Depth of analysis	No standard recipe exists. Every stage had to be designed, then verified against a reference or a known answer before the next stage could be trusted.
Familiarity of issues	Several pitfalls are not obvious: data leakage in cross-validation, artefacts that mimic the disease signal, confounding by sex, and metadata errors in public data.
Extent of applicable codes	Medical-device software regulation, data-protection law, the dataset licence, and research-ethics practice.
Interdependence	The reference choice changes the features, the cleaning changes the class balance of lost data, and the validation design decides whether the final number means anything.

2.4 Alternative solutions

- **Imaging and fluid biomarkers** (PET, MRI, CSF, blood tests): the most accurate, but expensive, centralised, and partly invasive. They are the confirmation step, not a screening step.

- **Cognitive questionnaires** such as the MMSE: cheap, but influenced by education and language, and insensitive early on.
- **Deep learning on raw EEG**: can learn its own features, but needs far more than 65 people, is hard to interpret, and is easy to inflate with epoch-level data leakage.
- **Connectivity and event-related EEG measures**: promising, but need more electrodes or task recordings.
- **Our choice: spectral DSP features with a simple, validated classifier**. It matches the known physiology, every step can be checked, the model can be explained, and it suits a small dataset.

2.5 Objectives and contributions

- Design and verify every DSP stage ourselves: filters, artefact removal, the Welch estimator and the features.
- Estimate honestly how well resting EEG separates AD from healthy people, using nested subject-wise cross-validation and a main analysis fixed in advance.
- Measure what each DSP step contributes (ablation) and what drives the result (feature families).
- Test whether the result survives other reasonable analysis choices and the age and sex imbalance.
- Report the limits openly: what the system cannot do (AD versus FTD), where it fails, and the data problem we found.

3 Design

This chapter formulates the problem and presents the design of every stage, with the theory behind it and the check that verified it.

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

Table 2. *Scope of the project*

In scope	Out of scope
Resting, eyes-closed, 19-channel EEG (OpenNeuro ds004504)	A clinical diagnosis: the output is screening support only
AD versus CN as the main question; FTD as a second question	Recording new data from patients
Designing, coding and verifying every DSP stage	Hardware or a real-time device
Testing on people never seen by the model	Deep learning (only 65 AD and CN people)
Alpha reactivity from the companion dataset ds006036	Disease staging or prediction of decline

3.1.2 Literature Review

EEG slowing in AD. The most replicated resting-EEG finding in AD is a shift of power from the alpha and beta bands to the theta and delta bands, together with a lower and slower posterior alpha peak and reduced reactivity of alpha to eye opening [3]. Ratios such as theta/alpha and slow/fast power summarise this shift in one number.

The dataset. Miltiadous *et al.* [4] published 88 eyes-closed resting recordings from a hospital in Greece (36 AD, 23 FTD, 29 CN) on OpenNeuro as ds004504, recorded with a Nihon Kohden EEG 2100 system at 500 Hz with 19 electrodes in the 10–20 layout and an A1–A2 mastoid reference. Diagnoses followed DSM and NINCDS–ADRDA criteria, and the MMSE was recorded. A companion dataset, ds006036 [5], contains eyes-open photic-stimulation recordings and states that its participant numbers match ds004504.

Validation pitfalls. Many EEG studies report accuracies above 95 % because epochs from the same person appear in both the training and the test data, so the model recognises the person rather than the disease [6]. Choosing features or tuning settings on all the data before cross-validation inflates performance in the same way, which nested cross-validation prevents [7]. Permutation tests give a p-value for the whole pipeline [8], confounds such as sex must be controlled inside cross-validation [9], and a multiverse or robustness analysis shows whether a result depends on arbitrary choices [10].

Methods we build on. Welch's averaged, windowed periodogram [11] is the standard low-variance spectral estimator; Hjorth parameters [12] give time-domain descriptors of the same spectral properties; independent component analysis separates blinks from brain activity [13], [14], [15]; and MNE-Python [16] and scikit-learn [17] provide tested building blocks that we used and verified.

The gap we address. For this dataset, published work usually reports only a final accuracy. We contribute a DSP chain in which every stage is verified, a main result declared before it was computed and tested against chance, an ablation that measures what each DSP step contributes, and robustness and confound analyses fixed in advance.

3.1.3 Formulation of Problem

Let $\mathbf{x}[n] \in \mathbb{R}^{19}$ be the multichannel EEG of one person, sampled at $f_s = 500$ Hz. The system is a mapping

$$\mathbf{x}[n] \in \mathbb{R}^{19} \rightarrow \hat{P}(f) \rightarrow \mathbf{z} \in \mathbb{R}^{94} \rightarrow p = P(\text{AD} | \mathbf{z}) \in [0,1] \quad (1)$$

Objective. Maximise the ability of p to rank AD patients above CN participants, measured by the AUC, on people never used for training or for choosing any setting.

Constraints. (i) No information from a test person may reach any step of training, including feature selection and tuning; (ii) no phase distortion, because the features compare bands; (iii) every stage must pass a built-in check against a known answer before it is used; (iv) with 65 AD and CN people, the model must be simple and strongly regularised.

Main analysis, declared before any result. AD versus CN; the 94 regional features; L2-regularised logistic regression with balanced class weights; AUC as the mean over 10 repeats of 5-fold cross-validation with seed 2026; a 95 % bootstrap confidence interval over people; and a 200-shuffle permutation test. Every other comparison, feature set and model is reported beside it and never swapped in as the headline.

3.1.4 Analysis

Each property of the signal was translated into a design specification before any design was attempted (Table 3).

Table 3. *From signal properties to design specifications*

Signal property	Requirement	Design decision
Brain rhythms of interest lie in 0.5–45 Hz	Remove drift below and muscle noise above	Band-pass 0.5–45 Hz
Strong 50 Hz mains interference	At least 40 dB rejection at 50 Hz	Notch, $Q = 30$
Features compare band powers and timing	No phase distortion	Zero phase (forward–backward)
Content ends at 45 Hz, recorded at 500 Hz	Keep a Nyquist margin, reduce load	Resample to 250 Hz
Theta (4–8 Hz) and alpha (8–13 Hz) are adjacent	Resolution 0.25 Hz or finer	4 s Welch segments
Blinks of hundreds of μV	Remove without discarding data	ICA + 150 μV rejection
Ear (A1–A2) reference adds a common signal	Make channels reflect local activity	Common-average reference

Where the filter specification comes from. The numbers were not chosen at random. The pass-band edges and the notch follow published clinical and research practice; the ripple, attenuation and transition values are engineering choices derived from the amplitudes involved and from the spectral resolution of the analysis. Each value, its reason and its source are given below.

- **High-pass edge, 0.5 Hz.** The clinical EEG guideline of the American Clinical Neurophysiology Society requires the low-frequency filter to be no higher than 1 Hz, so that slow activity is not attenuated [18]; 0.5 Hz satisfies this and keeps the whole delta band (0.5–4 Hz), which is where AD slowing appears. Below about 0.5 Hz the recording is dominated by electrode drift, sweating and movement. The dataset authors used the same lower edge [4].
- **Low-pass edge, 45 Hz.** The features use bands up to 45 Hz (gamma 25–45 Hz). Above this, scalp EEG is dominated by muscle activity and by the 50 Hz mains line, so the edge sits just below the line frequency. The dataset authors also band-limited their data to 0.5–45 Hz [4]. For visual reading of epileptic spikes, the clinical

guideline keeps 70 Hz [18]; that requirement protects sharp waveforms, whereas this project analyses band power, so the lower edge is appropriate.

- **50 Hz notch, $Q = 30$.** The mains frequency is 50 Hz where the data were recorded (Greece) and in Bangladesh. The band-pass alone attenuates 50 Hz by only 8 dB (forward–backward), so a notch is needed. $Q = 30$ gives a -3 dB width of $50/30 \approx 1.7$ Hz (about 48.3–51.7 Hz), narrow enough to leave the 45 Hz pass-band untouched. The guideline warns that notch filters can distort sharp transients [18]; this does not affect band-power features, and 50 Hz lies outside the analysed band.
- **Pass-band ripple ≤ 1 dB.** A conventional design tolerance [19]. It limits the gain error to at most 12 % in amplitude, and only in the frequency bins next to the band edges, which carry a small share of the band power.
- **Stop-band ≥ 60 dB (FIR) and ≥ 40 dB (IIR, forward–backward).** Slow drifts and artefacts can reach millivolts, while EEG rhythms are 10–100 μ V. 60 dB (a factor of 1000 in amplitude) brings a 1 mV drift down to 1 μ V, below the EEG background; for the Kaiser window this corresponds to $\beta = 5.653$ [20]. The IIR filter is applied twice, so it was designed for 20 dB per pass to reach 40 dB overall, the minimum accepted; it achieved 120.3 dB at 0.25 Hz.
- **Transition width 0.25 Hz at the low edge (0.25 $\rightarrow$ 0.5 Hz).** Equal to the frequency resolution of the Welch estimate ($\Delta f = 0.25$ Hz, 4 s segments): a sharper edge could not be seen in the spectrum, and a wider one would attenuate the lowest delta bin.
- **Upper stop-band at 60 Hz.** The 50 Hz line is handled by the notch, so the band-pass only needs to remove muscle activity above it. 60 Hz is also far below the 125 Hz Nyquist frequency after resampling to 250 Hz, so the band-pass doubles as protection against aliasing.
- **Zero phase, and IIR rather than FIR.** For offline analysis, zero-phase (forward–backward) filtering is recommended because it does not shift features in time; its non-causality only matters for analyses of onset latencies, which this project does not use [21]. The same guidance prefers FIR filters for their guaranteed stability and linear phase, and accepts IIR filters when stability is checked [21]. We designed both, verified that every IIR pole lies inside the unit circle (largest radius 0.9992), and used the IIR filter because it needs order 22 instead of 7253 taps [22].

The filter specification was written numerically: passband edges at 0.5 and 45 Hz no worse than -1 dB; stopband at 0.25 Hz and 60 Hz at least 60 dB down for the FIR design and at least 40 dB for the IIR design; and zero phase for both. Table 4 summarises the participants.

Table 4. *Participants of ds004504 (one FTD recording, sub-086, was later excluded)*

Group	n	Female / Male	Age (mean $\pm$ SD)	Age range	MMSE (mean $\pm$ SD)
AD	36	24 / 12	66.4 $\pm$ 7.9	49–79	17.8 $\pm$ 4.5
FTD	23	9 / 14	63.7 $\pm$ 8.2	44–78	22.2 $\pm$ 2.6
CN	29	11 / 18	67.9 $\pm$ 5.4	57–78	30.0 $\pm$ 0.0

Age is balanced between AD and CN ($p = 0.38$), but sex is not: 67 % of AD patients and 38 % of CN participants are women (Fisher's exact test, $p = 0.026$). This imbalance is addressed in Section 5.3.3.

3.2 Design Method (PO(a))

Fig. 3 shows the complete chain. Each stage below states the theory, the design, and the check that verified it.

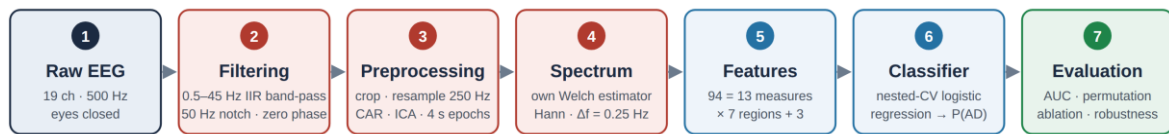


Figure 3. The designed processing chain, from one recording to a probability of AD

3.2.1 Data loading and channel layout

Each recording was read from its BIDS/EEGLAB files with MNE-Python [16]. The old 10–20 names T3, T4, T5 and T6 were renamed T7, T8, P7 and P8 so that the standard montage could assign 3-D electrode positions (Fig. 5). These positions are needed for topographic maps and for interpolating a faulty electrode. All 88 recordings were checked for duration, sampling rate and channel set before any processing.

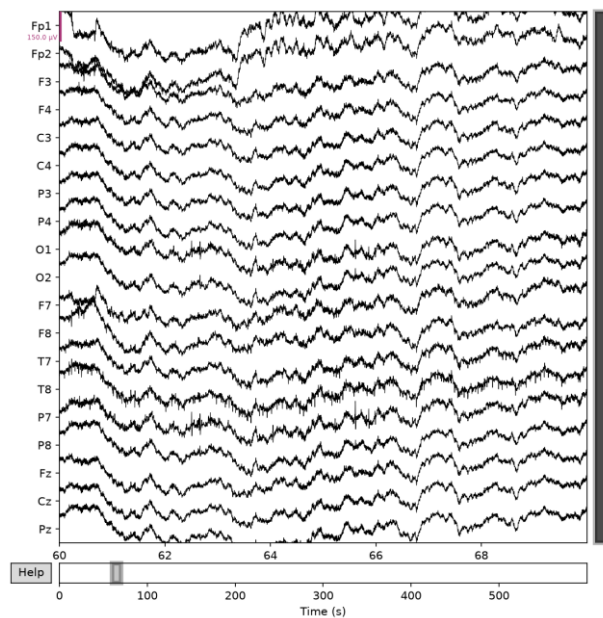


Figure 4. Ten seconds of one raw recording (19 channels) before any processing

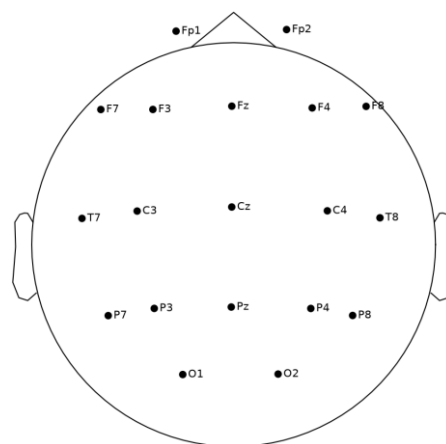


Figure 5. The 19 electrodes of the 10–20 layout used in ds004504

3.2.2 Filter design

FIR reference design. A linear-phase FIR band-pass was designed with the Kaiser window method. For a required stopband attenuation A (dB) and transition width Δf , Kaiser's formulas give the shape parameter and the length [20]:

$$\beta = 0.1102 (A - 8.7) \quad (A > 50), \quad N \approx \frac{A - 7.95}{2.285 \Delta\omega} + 1, \quad \Delta\omega = \frac{2\pi \Delta f}{f_s} \quad (2)$$

With $A = 60$ dB and a 0.25 Hz transition (0.25 $\rightarrow$ 0.5 Hz) at $f_s = 500$ Hz this gives $\beta = 5.653$ and $N = 7253$ taps. A window design places the cutoff at the -6 dB point in the middle of the transition band, so the cutoffs were set half a transition width outside the passband, at 0.375 Hz and 45.125 Hz; an earlier version with cutoffs at 0.5 and 45 Hz was 5.8 dB down at its own passband edges and was corrected. A Hamming design needed 6601 taps. The FIR filter has exactly linear phase with a delay of $(N - 1)/2 = 3626$ samples (7.25 s), which is removed by an index shift.

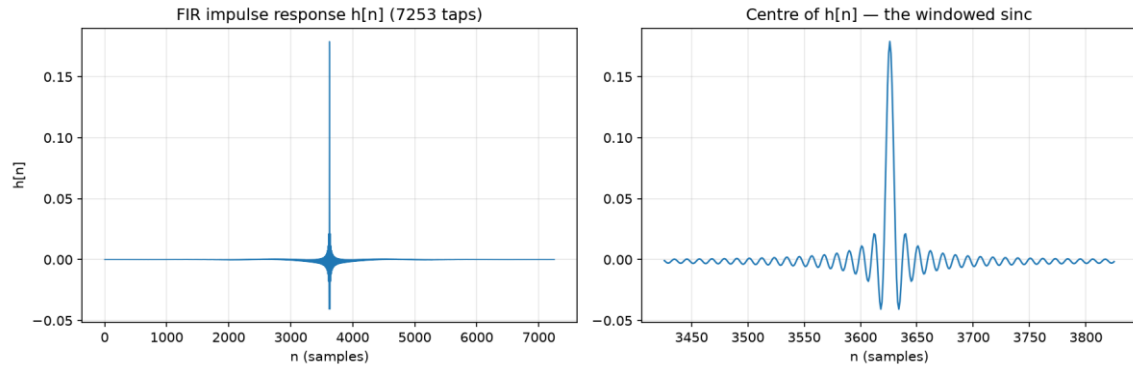


Figure 6. Impulse response of the 7253-tap Kaiser FIR band-pass (left) and its centre, the windowed sinc (right)

IIR design (used in the pipeline). A Butterworth band-pass has a maximally flat magnitude response [22]:

$$|H(j\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}} \quad (3)$$

The filter is always applied forward and backward (see zero-phase filtering below), which squares the magnitude and doubles every dB value. It was therefore designed to half the specification (0.5 dB ripple, 20 dB stopband) so that the forward–backward response lands on the full specification. The result is a prototype of order 11, a **band-pass of order 22**, realised as **11 second-order sections (SOS)**. Cascaded second-order sections avoid the coefficient sensitivity of a single high-order polynomial; the largest pole radius is 0.9992, so the filter is stable (Fig. 10). The IIR filter needs 22 poles where the FIR filter needs 7253 taps, and it ran in 0.09 s against 0.10 s for 19 channels $\times$ 300 s.

In transfer-function form the band-pass is a cascade of second-order sections,

$$H_{BP}(z) = \prod_{k=1}^{11} \frac{b_{0k} + b_{1k}z^{-1} + b_{2k}z^{-2}}{1 + a_{1k}z^{-1} + a_{2k}z^{-2}} \quad (4)$$

50 Hz notch. A second-order IIR notch with zeros on the unit circle at $\omega_0 = 2\pi \cdot 50/f_s$ and poles just inside them:

$$H_{\text{notch}}(z) = \frac{1 - 2\cos\omega_0 z^{-1} + z^{-2}}{1 - 2r\cos\omega_0 z^{-1} + r^2 z^{-2}}, \quad \omega_0 = \frac{2\pi \cdot 50}{f_s} \quad (5)$$

With $Q = 30$ the -3 dB bandwidth is $50/30 \approx 1.7$ Hz. The band-pass alone attenuates 50 Hz by only 8 dB forward–backward, so the notch is essential: the full chain gives -52.7 dB at 50 Hz in a single pass and about -105 dB forward–backward.

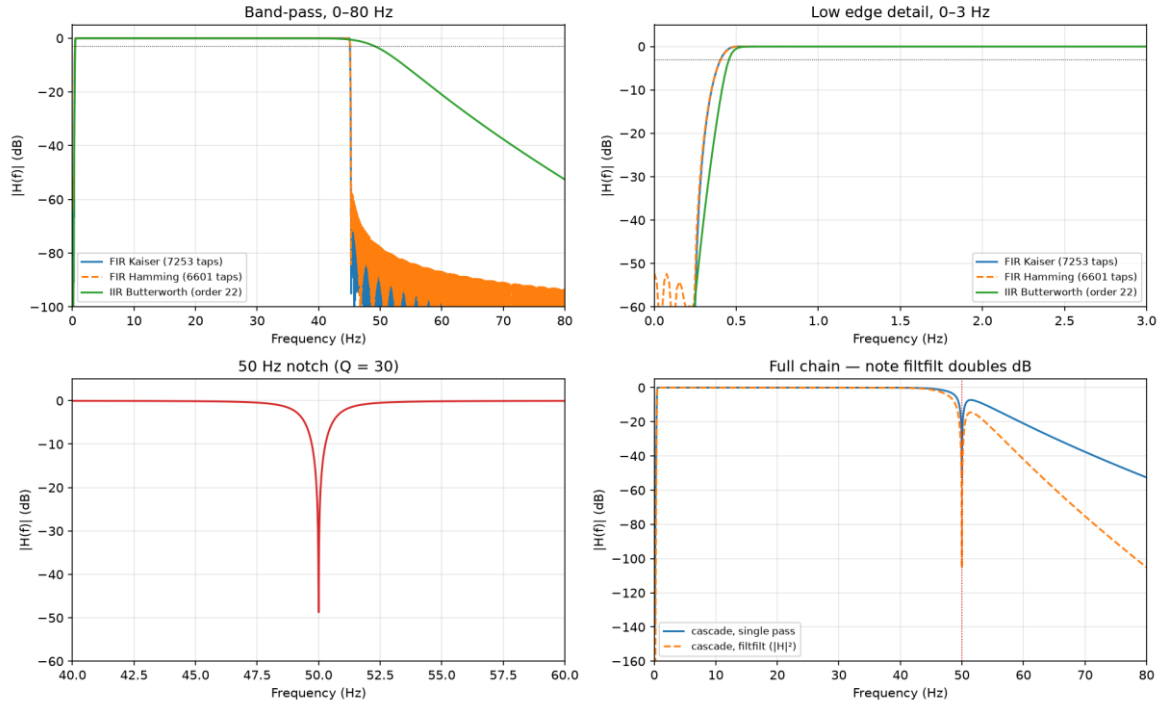


Figure 7. Magnitude responses: full band, low-edge detail, the 50 Hz notch and the full chain

Table 5. Both filter designs against the specification

Requirement	Specification	FIR (Kaiser)	IIR, forward–backward
Passband edges 0.5 / 45 Hz	no worse than –1 dB	0.0 / 0.0 dB ✓	–0.9 / –1.0 dB ✓
Stopband at 0.25 Hz	≥ 60 dB (FIR), ≥ 40 dB (IIR)	66.7 dB ✓	120.3 dB ✓
Stopband at 60 Hz	≥ 60 dB (FIR), ≥ 40 dB (IIR)	98.5 dB ✓	41.9 dB ✓
Phase	zero phase	delay removed by shift	forward–backward
Size	—	7253 taps	order 22 (11 SOS)

Zero-phase filtering. Applying $H(z)$ forward, reversing the output, filtering again and reversing back gives

$$Y(e^{j\omega}) = H(e^{j\omega}) H^*(e^{j\omega}) X(e^{j\omega}) = |H(e^{j\omega})|^2 X(e^{j\omega}) \quad (6)$$

so the phase cancels exactly and only the squared magnitude remains. This matters because a single IIR pass delays different frequencies by different amounts. The group delay of Eq. (7) of our band-pass is 152.7 ms at 2 Hz, 36.6 ms at 6 Hz, 27.2 ms at 10.5 Hz and 25.1 ms at 19 Hz, a 6.1-fold spread across the bands that the features compare (Fig. 9). Forward–backward filtering removes this distortion; it is possible because the analysis is offline. The 10 s cropped from each end of every recording (next section) is longer than the filters' edge transients.

$$\tau(\omega) = -\frac{d\phi(\omega)}{d\omega} \quad (7)$$

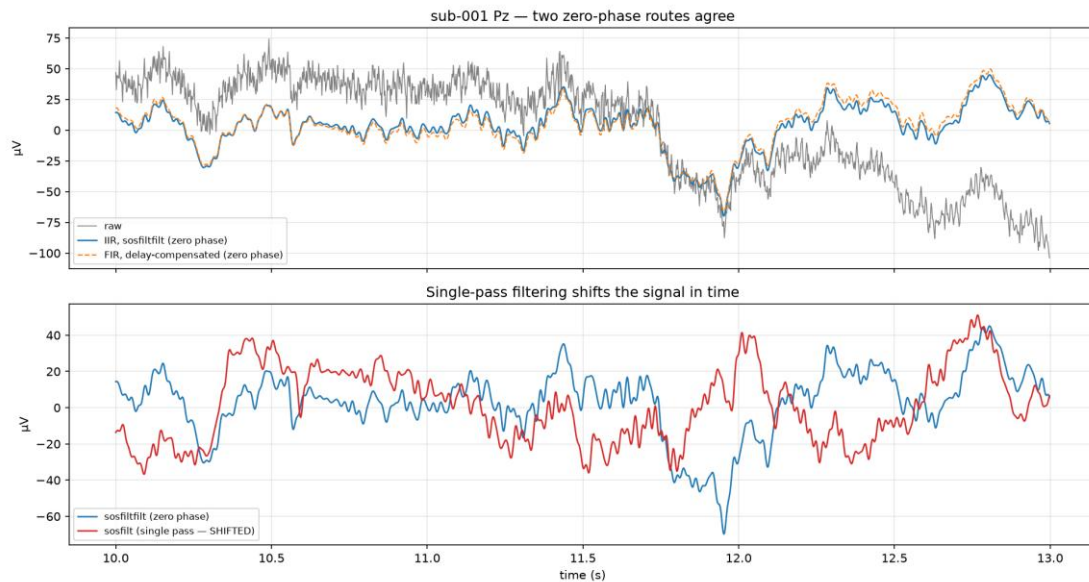


Figure 8. Channel Pz of sub-001: the two zero-phase routes agree (top), while a single IIR pass shifts the signal in time (bottom)

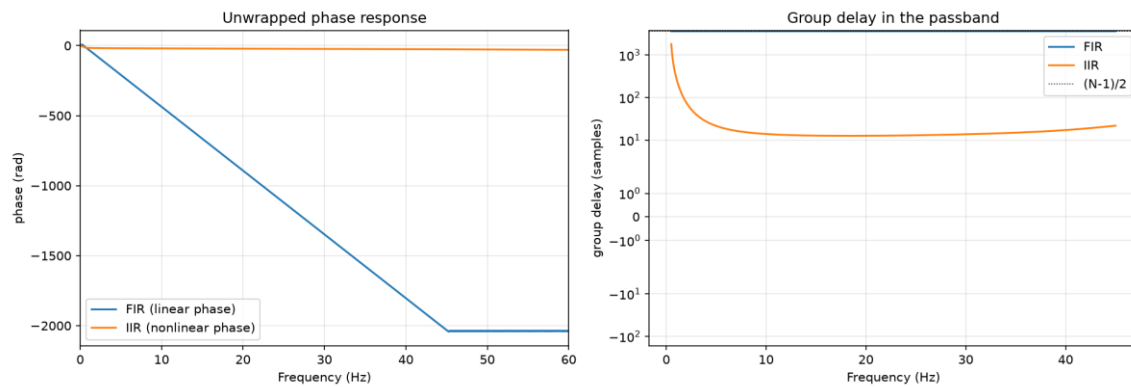


Figure 9. Phase and group delay of the single-pass designs; the IIR delay varies strongly across the passband

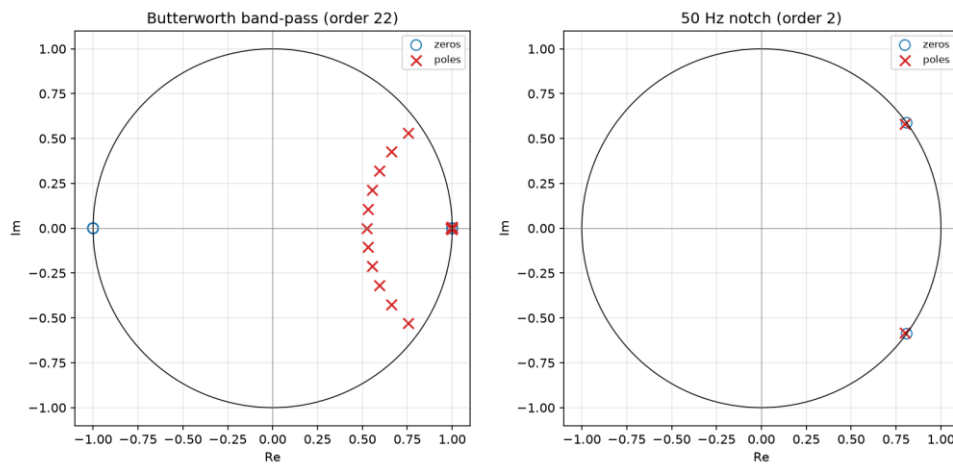


Figure 10. Pole-zero plots of the order-22 Butterworth band-pass and the 50 Hz notch; all poles lie inside the unit circle

3.2.3 Artefact removal and epoching

The cleaning chain was applied to the continuous recording in this order:

- **Notch and band-pass**, zero phase, as designed above.
- **Crop 10 s at each end** to remove filter edge transients.
- **Resample from 500 to 250 Hz**. The signal is band-limited to 45 Hz, so the new Nyquist frequency (125 Hz) leaves a wide margin; MNE's resampling applies its own anti-aliasing low-pass.
- **Repair faulty electrodes**. A channel was declared faulty if it exceeded the rejection limit in more than half of its epochs while no other channel did. Only sub-067 T8 met this rule (85 % of epochs, next-worst channel 22 %). It was rebuilt by spherical-spline interpolation [23] from the surrounding electrodes, before re-referencing, so its noise could not spread to other channels.
- **Common-average reference (CAR)**, defined below.
- **ICA** to remove the blink component, defined below.
- **Epoching and rejection**: 4 s epochs with 50 % overlap; any epoch with a peak-to-peak amplitude above 150 μV on any channel was discarded.

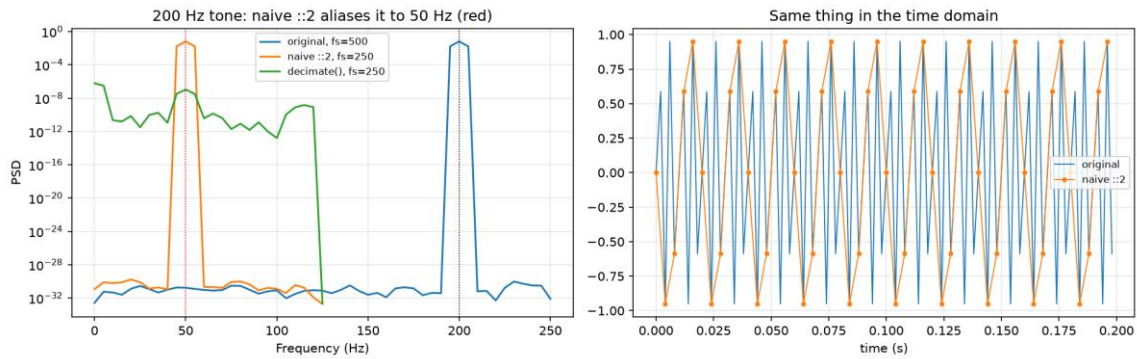


Figure 11. Why resampling needs an anti-aliasing filter: a 200 Hz tone decimated naively from 500 to 250 Hz aliases to 50 Hz (red), but not after low-pass filtering

Common-average reference. Every EEG voltage is a difference against a reference electrode. With the A1–A2 ear reference, any activity at the ears is subtracted into every channel equally. CAR subtracts the instantaneous mean of all $M = 19$ channels:

$$x_i^{\text{CAR}}[n] = x_i[n] - \frac{1}{M} \sum_{j=1}^M x_j[n] \quad (8)$$

A component common to all channels, such as the reference contribution, is also in the mean and cancels. After CAR the channels sum to zero, so the data rank falls from 19 to 18 (17 when an electrode has been interpolated).

Independent component analysis. Each electrode records a mixture of sources. With $\mathbf{s}$ the source signals and $\mathbf{A}$ the mixing weights, ICA estimates an unmixing matrix $\mathbf{W}$,

$$\mathbf{x}[n] = \mathbf{A} \mathbf{s}[n], \quad \hat{\mathbf{s}}[n] = \mathbf{W} \mathbf{x}[n] \quad (9)$$

so that the components $\hat{\mathbf{s}}$ are as statistically independent as possible [15]. Because a sum of independent signals is closer to Gaussian than its parts (central limit theorem), ICA searches for projections whose outputs are maximally non-Gaussian; extended Infomax [13], [14] handles both peaky (super-Gaussian) sources such as blinks and flat (sub-Gaussian) sources such as rhythms. We used MNE's implementation with 15 components (below the rank of 17–18, as a numerical margin and one setting for everybody), fitted on a 1 Hz high-passed copy of the data with every third sample and a fixed seed (97), then applied to the real data. ICA cannot label components, so the blink component was identified automatically by its correlation with Fp1 and Fp2, the electrodes nearest the

eyes (at most 3 components could be removed). In practice exactly one component was removed for every person. The channels were then rebuilt without it, so the blink is removed but the data are kept.

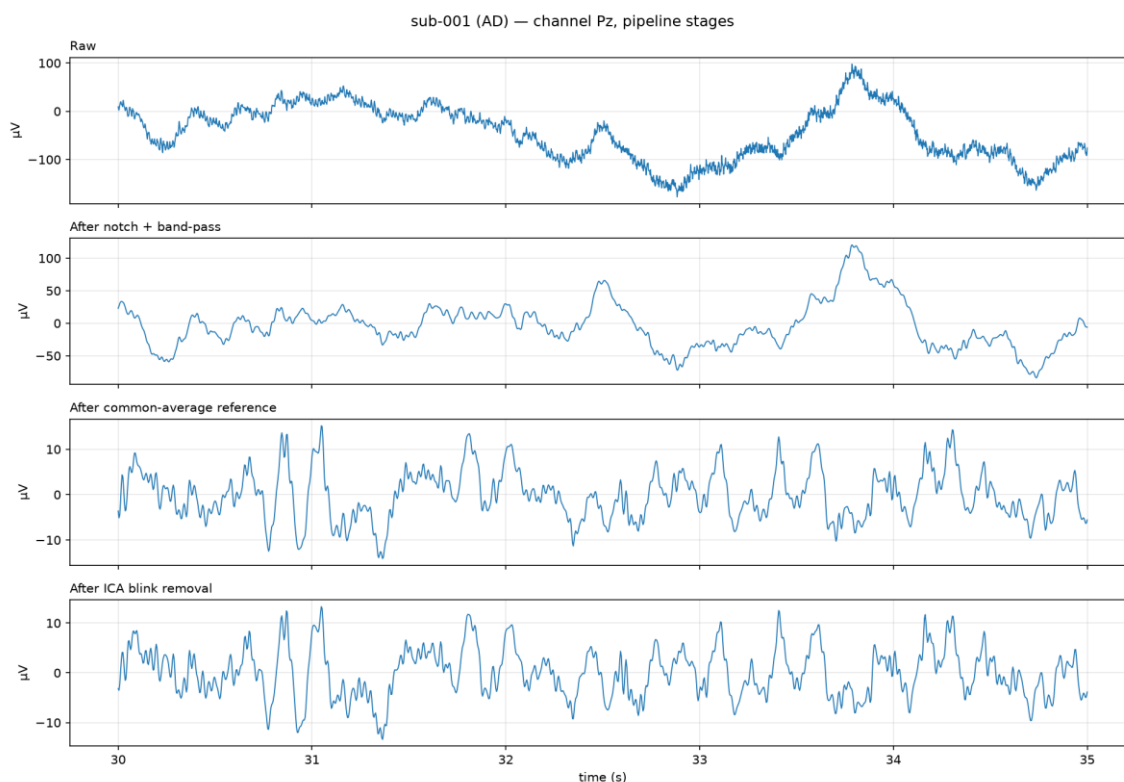


Figure 12. One AD patient (sub-001), channel Pz, after each stage of the pipeline

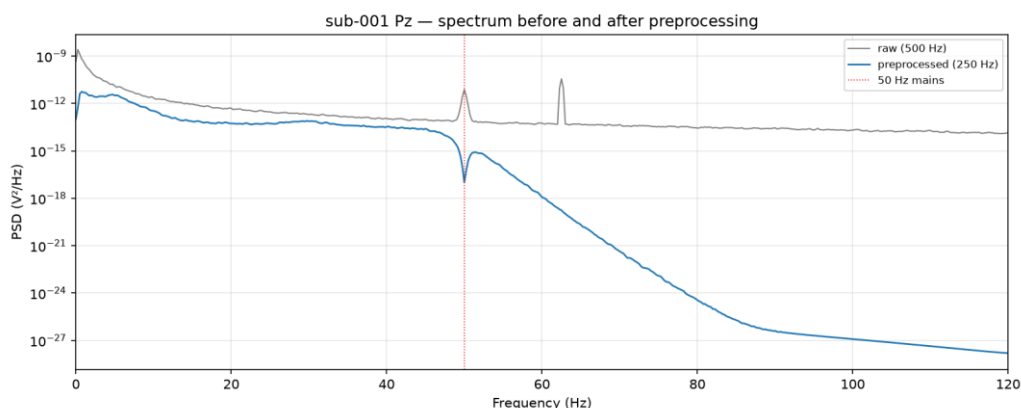


Figure 13. Spectrum of channel Pz of sub-001 before and after preprocessing: the 50 Hz line and the slow drift are removed

Why ICA matters for fairness. Without ICA, the 150 μV rule removed more data from AD patients than from CN participants (median 14.0 % against 8.7 %, Mann–Whitney $p = 0.076$). With ICA both groups lost about the same amount (5.3 % against 5.6 %, $p = 0.54$). After cleaning, the median number of 4 s epochs was 367 (AD), 382 (CN) and 320.5 (FTD); nobody had fewer than 20.

Exclusion. One FTD recording (sub-086) had five faulty electrodes and only 16 of 316 artefact-free epochs (under 2 minutes). A spectrum from 16 segments has about five times the relative error of one from about 370 segments (Section 3.2.4), and a quarter of its channels would have been interpolated, so it was excluded on data quality alone, before any classification. 87 recordings remain; the main AD versus CN analysis is unaffected.

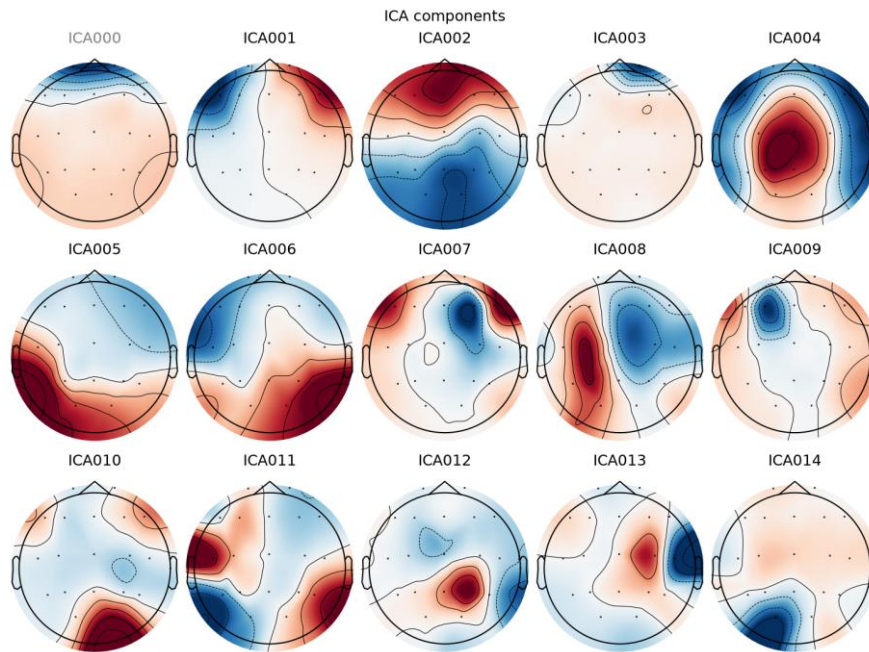


Figure 14. The 15 ICA components of sub-001; ICA000, a frontal blink component, was removed

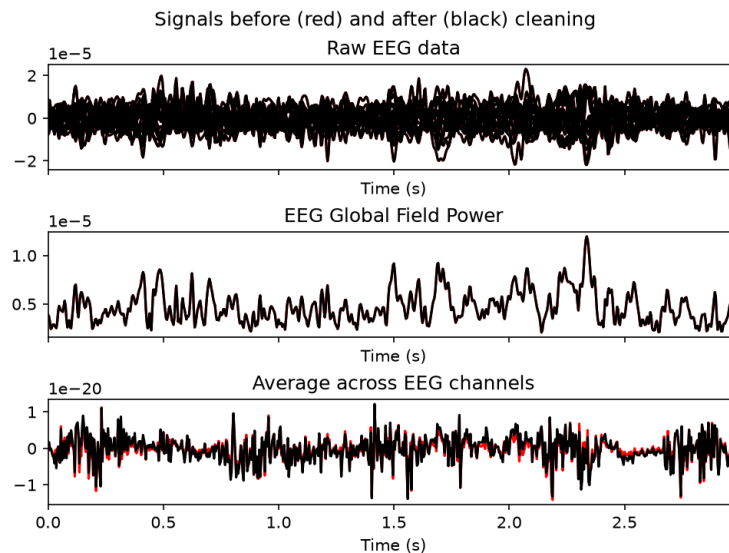


Figure 15. Signals before (red) and after (black) removal of the blink component, with the global field power

Check against the dataset authors' derivatives. For three test recordings our cleaned log-spectra correlated with the authors' published derivatives at $r = 0.977$ (AD), 0.988 (CN) and 0.978 (FTD). Our relative delta power was lower for the two patients. An ablation located the cause: replacing our filter with a conventional fourth-order Butterworth changed relative delta by less than 0.02, but disabling both CAR and ICA reproduced the authors' values to within 0.011. The difference therefore lies in artefact handling, not filtering: our pipeline removes slow, high-amplitude activity that the published derivatives retain. Because delta power can partly reflect eye movement, the theta/alpha ratio (which contains no delta) was chosen as the primary feature, and every feature was also computed over posterior electrodes only.

3.2.4 Spectral estimation: our own Welch estimator

A single periodogram of a random signal is an inconsistent estimator: at each frequency its standard deviation is about equal to its mean, however long the recording [11], [19]. A longer recording only gives finer frequency bins, each as noisy as before. Welch's method trades unneeded resolution for

averaging. The signal is cut into K segments $x_k[n]$ of length L , each multiplied by a window $w[n]$, and their periodograms are averaged:

$$\hat{P}(f) = \frac{1}{K} \sum_{k=1}^K \frac{|\sum_{n=0}^{L-1} w[n] x_k[n] e^{-j2\pi f n / f_s}|^2}{f_s \sum_{n=0}^{L-1} w^2[n]} \quad (10)$$

The denominator makes $\hat{P}$ a density in V^2/Hz ; the one-sided spectrum doubles every bin except DC and Nyquist. The frequency resolution is $\Delta f = f_s/L = 1/T$. With $T = 4$ s segments (our epochs), $\Delta f = 0.25$ Hz, fine enough to place a peak within the 8–13 Hz alpha band. A Hann window $w[n] = 0.5 - 0.5 \cos(2\pi n/(L-1))$ lowers the side lobes from -13 dB (rectangular) to -31 dB, so a strong alpha peak does not leak into neighbouring bands. For K roughly independent segments the relative standard deviation falls as $1/\sqrt{K}$; with about 370 segments per person, from 100 % to about 5 %:

$$w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{L-1}\right), \quad \Delta f = \frac{f_s}{L} = \frac{1}{T}, \quad \frac{\text{std}\{\hat{P}(f)\}}{\text{E}\{\hat{P}(f)\}} \approx \frac{1}{\sqrt{K}} \quad (11)$$

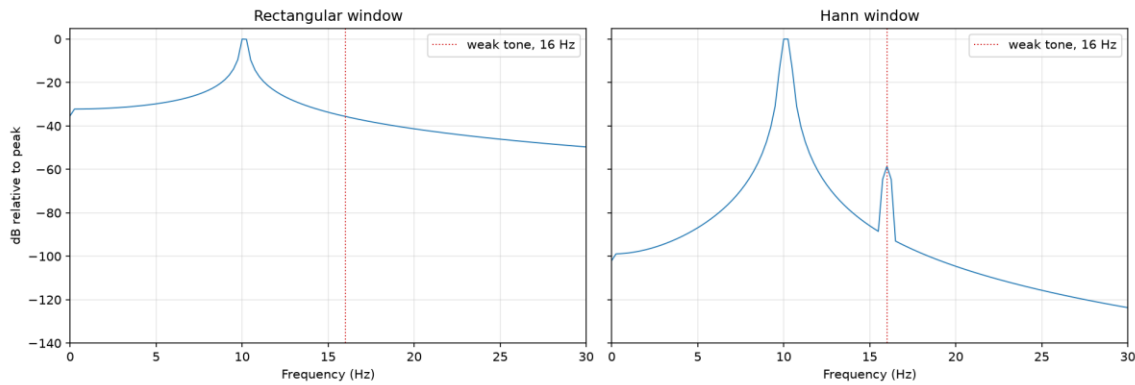


Figure 16. Spectral leakage: a strong 10.125 Hz tone hides a 16 Hz tone at -60 dB with a rectangular window (left) but not with the Hann window (right)

We implemented Eq. (10) in NumPy and validated it in five ways (Table 6, Fig. 17, Fig. 19).

Table 6. Validation of the Welch estimator implemented in this project

Check	Result
1. Against scipy.signal.welch, random data, five segment and overlap settings	relative error 1.2×10^{-15} to 2.8×10^{-15} (machine precision)
2. Against SciPy on real cleaned EEG (sub-001, 19 channels, 288 segments)	relative error $\approx 10^{-15}$
3. Epoch-based routine on the same segments	identical (error 0)
4. White noise, variance 1, $f_s = 250$ Hz	mean density 0.00797 (theory $2/f_s = 0.00800$); total power 0.9954 (theory 1)
5. 10 Hz sine of amplitude $10 \mu\text{V}$	power in 9–11 Hz = $5.0000 \times 10^{-11} \text{ V}^2$ (theory $A^2/2 = 5.0000 \times 10^{-11}$)
Leakage: 10.125 Hz tone, weak 16 Hz tone at -60 dB	rectangular window: level -35.6 dB (tone hidden); Hann: -58.6 dB (tone visible)
Scatter against Welch's theory on white noise	$K = 1$: 0.997 (theory 1.000); $K = 19$: 0.235 (0.235); $K = 299$: 0.062 (0.059)

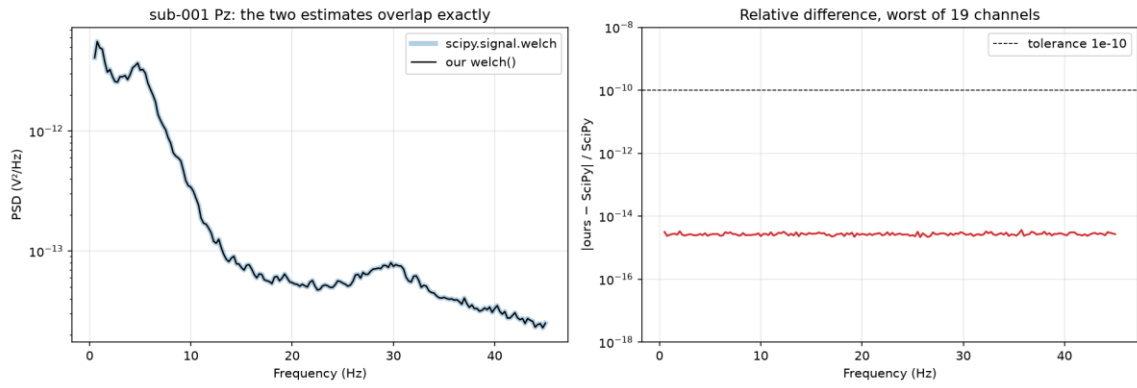


Figure 17. Our estimate and SciPy's for channel Pz of sub-001 (left) and the relative difference over 19 channels (right), far below the 10^{-10} tolerance

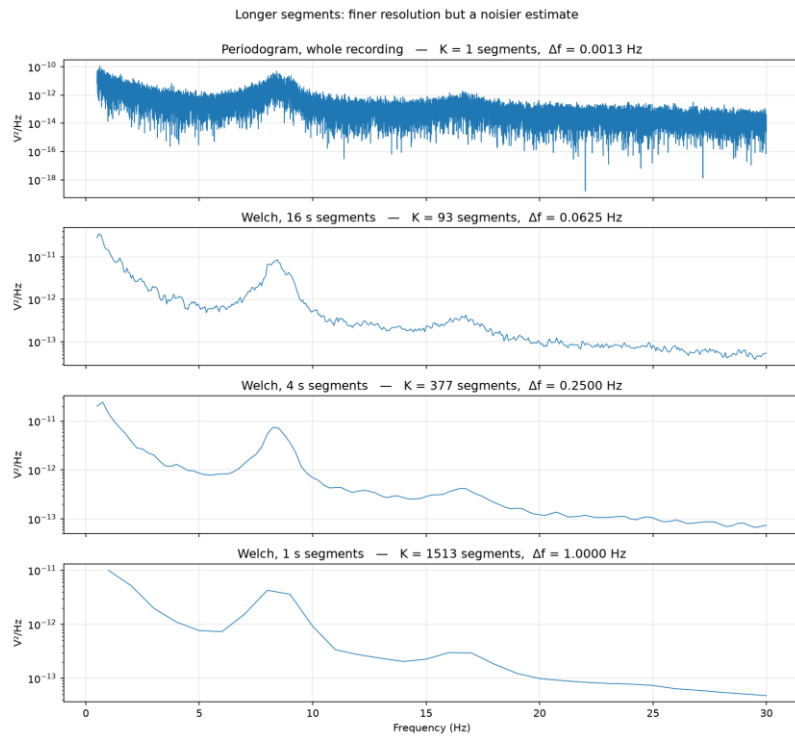


Figure 18. The same recording estimated four ways: a single periodogram is a noisy band; 4 s Welch segments give a smooth spectrum with a sharp alpha peak

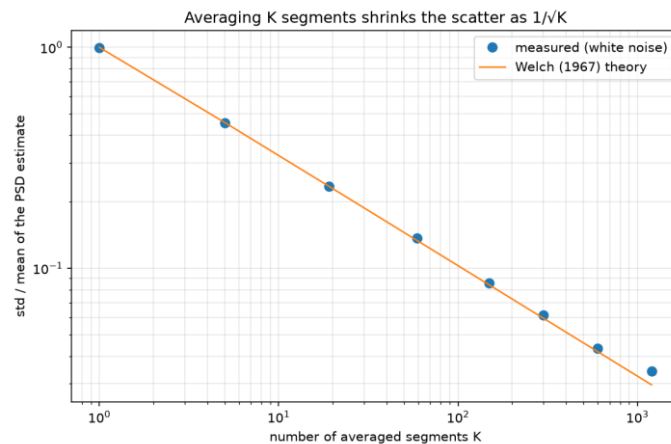


Figure 19. Measured relative scatter of the estimate against the number of averaged segments, following Welch's $1/\sqrt{K}$ law

Averaging segments in time is equivalent to pooling neighbouring bins of one long periodogram; Welch's form was chosen because it also works around the gaps left by rejected epochs, its windowing is standard, and it could be verified exactly against a reference.

3.2.5 Feature extraction

From each channel's spectrum (0.5–45 Hz) and from the clean epochs, 13 measures were computed (Table 7). Band powers were integrated with the trapezoid rule and divided by the 0.5–45 Hz total, so the five relative powers sum to one. Each measure was then averaged over seven regions: global (all 19), frontal (Fp1, Fp2, F7, F3, Fz, F4, F8), central (C3, Cz, C4), temporal (T7, T8, P7, P8), parietal (P3, Pz, P4), occipital (O1, O2) and posterior (P3, P4, Pz, O1, O2, P7, P8). Ratios were averaged as logarithms, which limits the pull of extreme electrodes.

For a band b with power P_b and the spectrum normalised to a distribution $p(f)$, the main definitions are

$$P_b = \frac{\int_{f \in b} \hat{P}(f) df}{\int_{0.5}^{45} \hat{P}(f) df}, \quad \log\text{TAR} = \log_{10} \frac{P_\theta}{P_\alpha}, \quad H_s = -\frac{1}{\ln N_f} \sum_f p(f) \ln p(f) \quad (12)$$

$$\text{Activity} = \sigma_x^2, \quad \text{Mobility} = \frac{\sigma_{x'}}{\sigma_x}, \quad \text{Complexity} = \frac{\text{Mobility}(x')}{\text{Mobility}(x)} \quad (13)$$

Table 7. The 94 regional features (a channel-level set of $13 \times 19 + 3 = 250$ features was also built)

Family	Measure and definition	Features
Band powers	Relative power in δ 0.5–4, θ 4–8, α 8–13, β 13–25, γ 25–45 Hz	$5 \times 7 = 35$
Ratios	$\log\text{TAR} = \log_{10}(P_\theta/P_\alpha)$; $\log\text{SlowFast} = \log_{10}((P_\delta + P_\theta)/(P_\alpha + P_\beta))$	$2 \times 7 = 14$
Spectral shape	Spectral entropy $-\sum p(f) \ln p(f) / \ln N_f$ (p = normalised PSD, 0 for a pure tone, 1 for white noise); SEF50 and SEF95: frequencies below which 50 % and 95 % of power lie	$3 \times 7 = 21$
Hjorth	Activity $\text{var}(x)$ (as log); mobility $\sqrt{(\text{var}(x')/\text{var}(x))}$; complexity $\text{mobility}(x')/\text{mobility}(x)$	$3 \times 7 = 21$
Alpha peak	Posterior 1/f background fitted (2–40 Hz) and removed; peak present if it rises more than 0.1 $\log_{10}$ units (26 %) above it within 6–14 Hz; its frequency and height	3
Total		94

The alpha-peak feature replaced an earlier, naive definition (the highest point between 6 and 13 Hz). Because EEG power falls with frequency, a person without any alpha peak was assigned the window floor of 6.0 Hz; this happened for 11 of 36 AD patients and none of 29 CN participants (Fisher's exact test, $p = 0.0007$). Fitting and removing the 1/f background first, and recording whether a peak exists, turns that artefact into a meaningful feature.

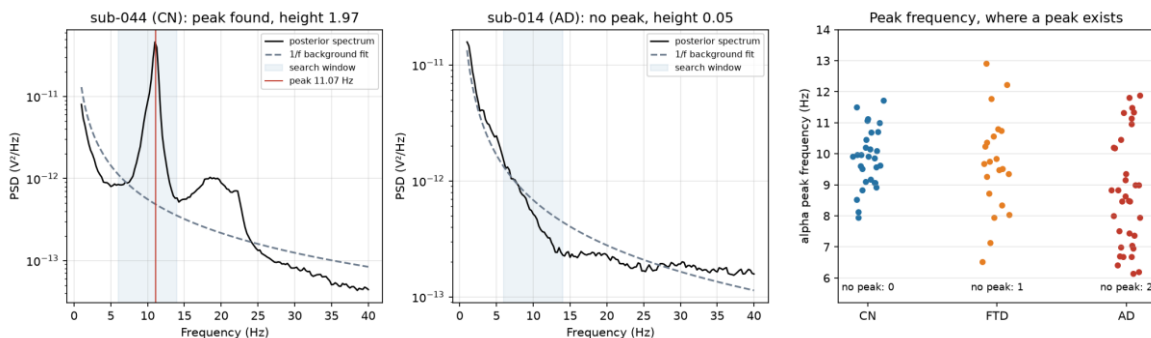


Figure 20. Alpha-peak detection after removing the 1/f background: a clear peak in a CN participant (left), no peak in an AD patient (centre), and peak frequencies where a peak exists (right)

Fig. 21 shows where on the scalp single features separate AD from CN, and Fig. 22 shows the strongest features. The theta/alpha ratio is high in AD in every region (single-feature AUC 0.872–0.884), with FTD in between. These single-feature AUCs use all participants and pick the best of many, so they are descriptive only; the honest estimate comes from the cross-validated classifier.

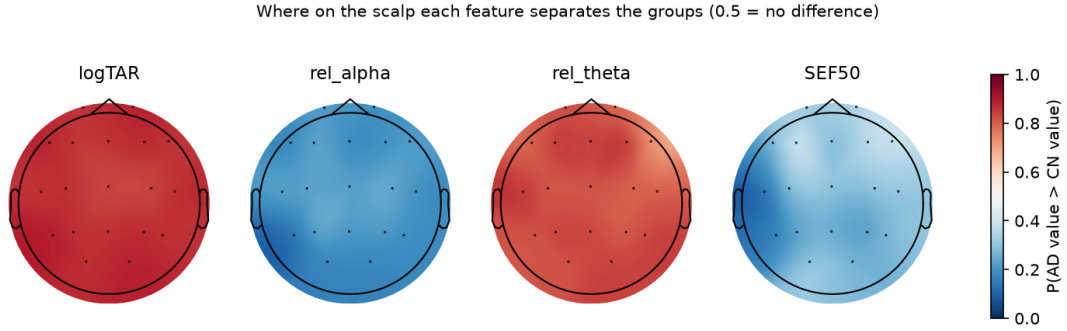


Figure 21. Single-feature AUC for AD against CN at every electrode (red: higher in AD, blue: lower in AD)

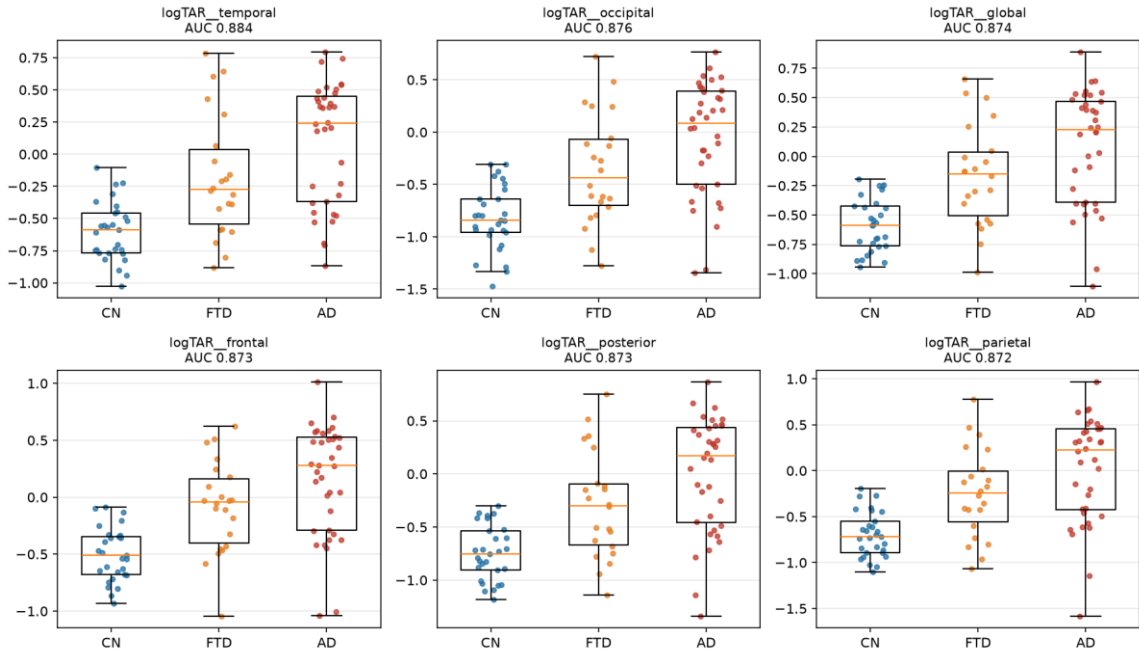


Figure 22. The six regional theta/alpha ratios by group; CN low, AD high, FTD in between

3.2.6 Classification and honest validation

Model. Logistic regression combines the features into one probability:

$$p(\text{AD} \mid \mathbf{z}) = \sigma(\mathbf{w}^\top \mathbf{z} + b) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{z} + b)}} \quad (14)$$

The weights minimise the class-balanced logistic loss plus an L2 penalty,

$$\min_{\mathbf{w}, b} \frac{\|\mathbf{w}\|^2}{2C} - \sum_i c_{y_i} [y_i \ln p_i + (1 - y_i) \ln(1 - p_i)] \quad (15)$$

where c_y weights each class inversely to its size. The model was chosen before any result for four reasons: with 65 people and 94 features a linear, regularised model is least prone to overfitting; the expected relationship (more slowing, more likely AD) is monotonic; it outputs probabilities directly; and its weights are interpretable. Each fitted pipeline is: median imputation (the alpha-peak frequency is undefined when no peak exists) → standardisation → selection of the k best features by an ANOVA F-test → logistic regression. $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ and $k \in \{10, 25, 50, \text{all}\}$ were tuned.

The comparison models (linear and RBF support-vector machines, a 200-tree random forest and k-nearest neighbours) used the same pipeline with their own grids.

Nested cross-validation. The outer loop splits the 65 people into 5 stratified folds of 13, trains on 52 and tests on the remaining 13, and rotates so that every person is tested once by a model that never saw them; the whole procedure is repeated 10 times with different splits. Inside every outer training set, an inner 5-fold loop chooses C and k using those 52 people only. Imputation, scaling and feature selection are fitted inside each training set as part of the pipeline. Leave-one-subject-out was rejected because it is biased for AUC: a model that only learns the class ratio scored AUC 0.000 with leave-one-out but 0.475 with 5-fold cross-validation.

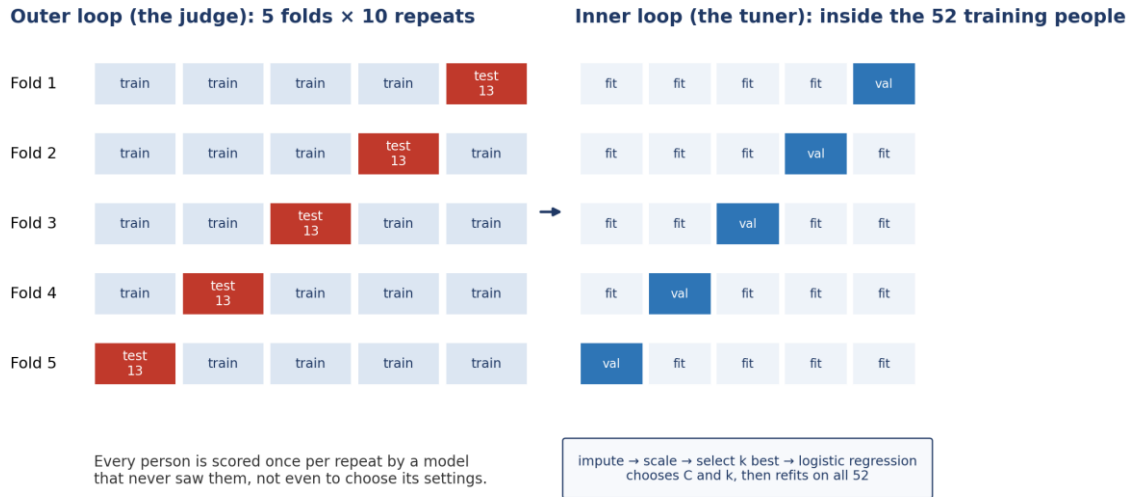


Figure 23. Nested cross-validation: the outer loop scores people never seen by the model; the inner loop chooses the settings inside the training people only

Scores and uncertainty. The AUC equals the probability that a randomly chosen AD patient receives a higher score than a randomly chosen CN participant (the Mann–Whitney statistic [24]). It was computed from the pooled out-of-sample probabilities of each repeat and averaged over the 10 repeats:

$$\text{AUC} = \frac{1}{n_1 n_0} \sum_{i \in \text{AD}} \sum_{j \in \text{CN}} [\mathbb{1}(p_i > p_j) + 1/2 \mathbb{1}(p_i = p_j)], \quad \text{BA} = \frac{\text{Se} + \text{Sp}}{2} \quad (16)$$

Balanced accuracy, the mean of sensitivity and specificity at $p = 0.5$, is reported beside it. 95 % confidence intervals come from 2000 stratified bootstrap resamples of people [25]. The permutation test shuffled the diagnoses 200 times and reran the entire nested pipeline each time (with 2 outer repeats). With b the number of the $B = 200$ shuffles that reached the real score [8],

$$p = \frac{b + 1}{B + 1} \quad (17)$$

Checking the checker. Before any real result, the validation machinery was run on data with a known answer (Fig. 24). Pure noise features gave AUC 0.476 and shuffled labels 0.506 (both chance, as required); a planted 1-SD difference in 5 of 94 noise features gave 0.801 (the signal is found). The decisive test was the feature-selection trap: on pure noise, choosing the 10 best features using all people and then cross-validating produced a fake AUC of 0.779, while choosing them inside each training fold gave 0.459, the correct answer. This is why every choice in the pipeline is made inside the training data.

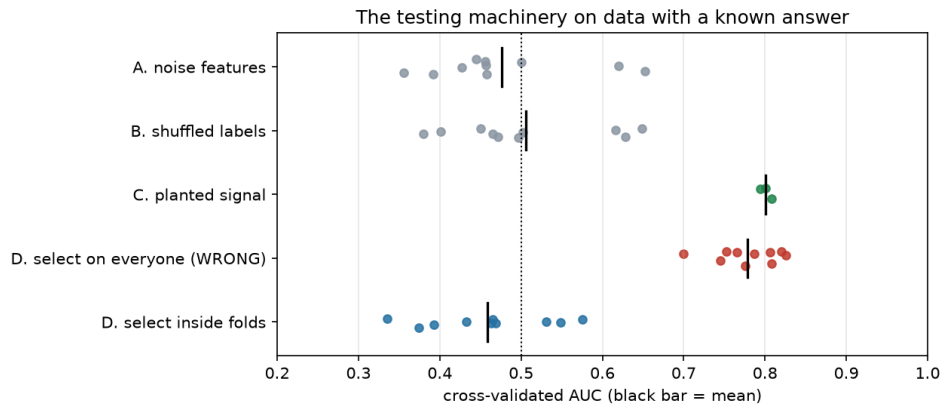


Figure 24. The validation machinery on data with a known answer; only selecting features on everyone (red) produces a falsely high AUC

FINAL DESIGN AT A GLANCE

- **Filters:** IIR Butterworth band-pass 0.5–45 Hz, order 22 (11 SOS), plus a 50 Hz notch ($Q = 30$), applied forward–backward (zero phase).
- **Cleaning:** crop 10 s per end; resample to 250 Hz; spherical-spline repair of faulty electrodes; common-average reference; extended-Infomax ICA (15 components, blink removed); 4 s epochs with 50 % overlap; 150 μ V rejection.
- **Spectrum:** own Welch estimator, Hann window, 4 s segments, $\Delta f = 0.25$ Hz, about 370 segments per person.
- **Features:** 13 measures $\times$ 7 regions + 3 alpha-peak features = 94.
- **Model and validation:** L2 logistic regression with balanced weights; nested 5×10 cross-validation; bootstrap CIs; 200-shuffle permutation test.

3.3 Simulation Model

This project has no circuit, but its design was simulated in two ways before and alongside the real analysis.

Synthetic signals with known answers. Every DSP stage was first exercised on signals whose correct output is known: white noise of unit variance and a 10 μ V, 10 Hz sine for the Welch estimator (Table 6), a strong and a weak tone for spectral leakage, and synthetic multichannel data for the whole preprocessing chain. The classification machinery was simulated on pure noise, shuffled labels and a planted signal (Fig. 24). A stage was used on real data only after its simulation passed.

Design ablation: does each DSP step matter? The same 94 features and the same classifier were rebuilt from the raw recordings with more and more of the designed processing switched on (Table 8, Fig. 25).

Table 8. Cleaning ablation with identical features and classifier

Processing level	AD vs CN AUC	FTD vs CN AUC
L0: raw data, no filter	0.725	0.613
L1: + our band-pass and notch	0.717	0.678
L2: + common-average reference	0.847	0.703
L2r: + 150 μ V rejection (no ICA)	0.795	0.751
L3: + ICA (full pipeline)	0.811	0.728

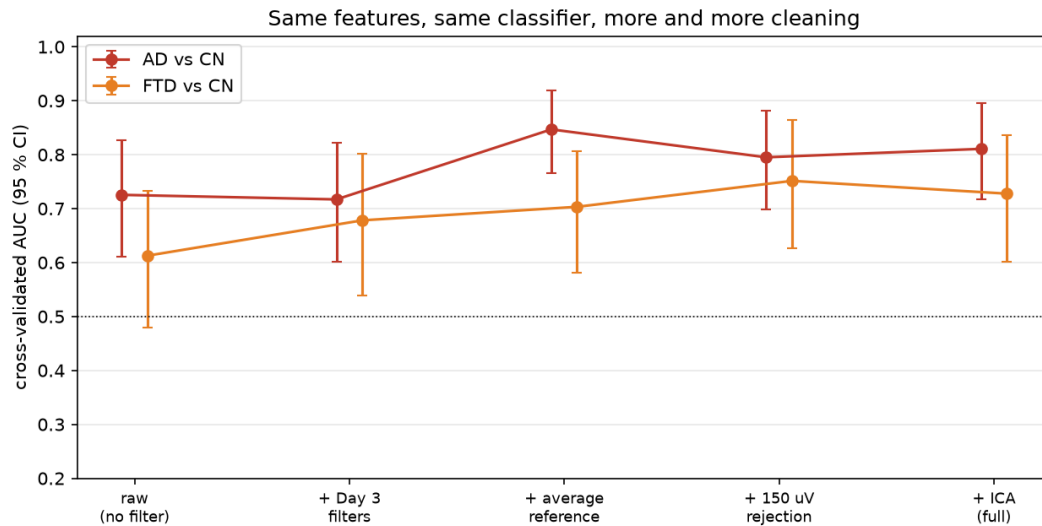


Figure 25. AUC with 95 % intervals at each processing level

Three conclusions follow. First, processing matters: the full pipeline beats raw data by 0.085 for AD ($p = 0.084$) and by 0.115 for FTD ($p = 0.027$). Second, the **single most important DSP step is the common-average reference** (0.717 $\rightarrow$ 0.847 for AD): the ear reference adds one shared rhythm to every channel and makes all regions look alike. Filtering alone changes little for AD, because the features already use only 0.5–45 Hz and rely on ratios. Third, the uncleaned level L2 scores higher than the full pipeline (0.847 against 0.811, difference not significant). Eye movements add slow delta and theta power that resembles AD slowing, and patients produced more of them; a higher score for the wrong reason is not a result. The report therefore keeps 0.811, the score of the pipeline that was designed and declared in advance.

3.4 Full Source Code

The five core modules that implement the design are listed below with comments and docstrings removed, in two columns (read the left column, then the right). The data-access module, the demonstration tool, the analysis scripts and the commented originals are in the project repository. The code requires Python 3 with NumPy, SciPy, pandas, matplotlib, MNE-Python and scikit-learn.

Listing 1. filters.py: filter design and zero-phase application

```
import numpy as np
from scipy import signal

FS = 500.0
L_FREQ = 0.5
H_FREQ = 45.0
LINE_FREQ = 50.0

def design_fir_bandpass(fs=FS, l_freq=L_FREQ,
                        h_freq=H_FREQ,
                        trans_width=0.25,
                        ripple_db=60.0):
    nyq = fs / 2.0
    numtaps, beta = signal.kaiserord(ripple_db,
    trans_width / nyq)
    if numtaps % 2 == 0:
        numtaps += 1
    taps = signal.firwin(numtaps,
    [l_freq - trans_width / 2.0,
    h_freq + trans_width / 2.0],
    window="kaiser", beta),
    pass_zero=False, fs=fs)
    return taps, beta

def design_fir_hamming(fs=FS, l_freq=L_FREQ,
                        h_freq=H_FREQ,
                        trans_width=0.25):
    numtaps = int(np.ceil(3.3 * fs / trans_width))
```

```
if numtaps % 2 == 0:
    numtaps += 1
    taps = signal.firwin(numtaps,
    [l_freq - trans_width / 2.0,
    h_freq + trans_width / 2.0],
    window="hamming",
    pass_zero=False, fs=fs)
    return taps

def design_iir_bandpass(fs=FS, l_freq=L_FREQ,
                        h_freq=H_FREQ,
                        gpass=0.5, gstop=20.0):
    wp = [l_freq, h_freq]
    ws = [l_freq / 2.0, 60.0]
    N, Wn = signal.buttord(wp, ws, gpass, gstop, fs=fs)
    sos = signal.butter(N, Wn, btype="bandpass",
    output="sos", fs=fs)
    return sos, N

def design_notch(fs=FS, f0=LINE_FREQ, Q=30.0):
    b, a = signal.iirnotch(f0, Q, fs=fs)
    return signal.tf2sos(b, a)

def sos_response(sos, fs=FS, worN=16384):
    w, h = signal.sosfreqz(sos, worN=worN, fs=fs)
    return w, h

def group_delay_samples(w_hz, h, fs=FS):
    phase = np.unwrap(np.angle(h))
    return -np.gradient(phase, w_hz) * fs / (2.0 *
    np.pi)
```

```

def db(h, floor=-140.0):
    mag = np.abs(h)
    out = 20.0 * np.log10(np.maximum(mag, 1e-12))
    return np.maximum(out, floor)

def apply_fir_zerophase(x, taps):
    x = np.atleast_2d(x)
    d = (len(taps) - 1) // 2
    y = signal.oaconvolve(x, taps[None, :], mode="full",
        axes=-1)
    return y[:, d:d + x.shape[-1]]

def apply_iir_zerophase(x, sos):
    return signal.sosfiltfilt(sos, np.atleast_2d(x),
        axis=-1)

def preprocess_array(x, fs=FS, use_notch=True):
    sos_bp, _ = design_iir_bandpass(fs=fs)
    y = np.atleast_2d(x)
    if use_notch:
        y = signal.sosfiltfilt(design_notch(fs=fs), y,
            axis=-1)
    return signal.sosfiltfilt(sos_bp, y, axis=-1)

```

Listing 2. spectral.py: our Welch estimator

```

import numpy as np

TRAPZ = getattr(np, "trapezoid", None) or np.trapz

def hann(n):
    k = np.arange(n)
    return 0.5 - 0.5 * np.cos(2.0 * np.pi * k / n)

def n_segments(n_samples, nperseg, noverlap):
    step = nperseg - noverlap
    return (n_samples - noverlap) // step

def segment(x, nperseg, noverlap):
    step = nperseg - noverlap
    k = n_segments(x.shape[-1], nperseg, noverlap)
    if k < 1:
        raise ValueError("signal is shorter than one segment")
    idx = np.arange(nperseg)[None, :] + step *
        np.arange(k)[:, None]
    return x[..., idx]

def _segment_spectra(seg, fs, window, detrend=True):
    n = seg.shape[-1]
    if detrend:
        seg = seg - seg.mean(axis=-1, keepdims=True)

    spec = np.fft.rfft(seg * window, axis=-1)
    p = (spec.real ** 2 + spec.imag ** 2) / (fs *
        np.sum(window ** 2))

    if n % 2 == 0:
        p[..., 1:-1] *= 2.0
    else:
        p[..., 1:] *= 2.0

    freqs = np.fft.rfftfreq(n, d=1.0 / fs)
    return freqs, p

def welch(x, fs, nperseg, noverlap=None, window=None,
    detrend=True):
    x = np.asarray(x, dtype=float)
    if noverlap is None:
        noverlap = nperseg // 2

    w = hann(nperseg) if window is None else
        np.asarray(window, dtype=float)
    seg = segment(x, nperseg, noverlap)
    freqs, p = _segment_spectra(seg, fs, w, detrend)

    return freqs, p.mean(axis=-2)

def welch_epochs(data, fs, window=None, detrend=True):
    data = np.asarray(data, dtype=float)
    n = data.shape[-1]
    w = hann(n) if window is None else
        np.asarray(window, dtype=float)
    freqs, p = _segment_spectra(data, fs, w, detrend)
    return freqs, p.mean(axis=0)

def welch_variance_ratio(nperseg, noverlap, k,
    window=None):
    w = hann(nperseg) if window is None else
        np.asarray(window, dtype=float)

```

```

    d = nperseg - noverlap
    energy = np.sum(w ** 2)
    total = 1.0
    j = 1
    while j < k and j * d < nperseg:
        rho = (np.sum(w[: nperseg - j * d] * w[j * d:]))
        / energy) ** 2
        total += 2.0 * (k - j) / k * rho
        j += 1
    return total / k

def band_power(freqs, psd, lo, hi):
    m = (freqs >= lo) & (freqs <= hi)
    return TRAPZ(psd[..., m], freqs[m], axis=-1)

```

Listing 3. preprocess.py: the cleaning pipeline

```

from __future__ import annotations

import sys
from dataclasses import dataclass, field
from pathlib import Path

import mne
import numpy as np
from scipy import signal

sys.path.append(str(Path(__file__).resolve().parent))
import filters as F
from eegio import DS_REST, load_raw

CROP_EDGE_S = 10.0
TARGET_SFREQ = 250.0
ICA_METHOD = "infomax"
ICA_N_COMPONENTS = 15
ICA_HIGHPASS = 1.0
ICA_RANDOM_STATE = 97
ICA_DECIM = 3
ICA_MAX_BAD = 3
EOG_PROXY = ["Fp1", "Fp2"]
EPOCH_LEN_S = 4.0
EPOCH_OVERLAP_S = 2.0
REJECT_PTP = 150e-6

BAD_CHANNELS = {
    "sub-067": ["T8"],
}

@dataclass
class PreprocReport:
    subject: str
    n_channels: int = 0
    dur_raw_s: float = 0.0
    dur_cropped_s: float = 0.0
    sfreq_out: float = 0.0
    ica_method_used: str = ""
    ica_excluded: list = field(default_factory=list)
    n_epochs_made: int = 0
    n_epochs_kept: int = 0
    interpolated: list = field(default_factory=list)

    @property
    def drop_pct(self) -> float:
        if not self.n_epochs_made:
            return float("nan")
        return 100.0 * (1.0 - self.n_epochs_kept /
            self.n_epochs_made)

def _set_filter_info(raw, l_freq, h_freq):
    try:
        with raw.info.unlock():
            raw.info["highpass"] = float(l_freq)
            raw.info["lowpass"] = float(h_freq)
    except Exception:
        pass

def apply_own_filters(raw):
    fs = raw.info["sfreq"]
    sos_notch = F.design_notch(fs=fs)
    sos_bp, _ = F.design_iir_bandpass(fs=fs)

    def _fun(data):
        y = signal.sosfiltfilt(sos_notch, data, axis=-1)
        return signal.sosfiltfilt(sos_bp, y, axis=-1)

    raw.apply_function(_fun, picks="eeg",
        channel_wise=False)
    _set_filter_info(raw, F.L_FREQ, F.H_FREQ)
    return raw

def _normalise_scores(scores):

```

```

arr = np.asarray(scores, dtype=float)
if arr.ndim == 2:
    arr = np.max(np.abs(arr), axis=0)
return np.abs(arr)

def detect_blink_components(ica, inst, report):
    inds, scores = [], None
    try:
        inds, raw_scores = ica.find_bads_eog(
            inst, ch_name=EKG_PROXY, threshold=3.0,
            verbose="ERROR")
        scores = _normalise_scores(raw_scores)
        report.ica_method_used = "find_bads_eog"
    except Exception as exc:
        print(f"find_bads_eog failed
        ({type(exc).__name__}); "
            f"using correlation fallback")

    if not inds:
        src = ica.get_sources(inst).get_data()
        proxy =
inst.get_data(picks=EKG_PROXY).mean(axis=0)
        sos = signal.butter(4, [1.0, 10.0],
            btype="band",
            output="sos",
            fs=inst.info["sfreq"])
        proxy = signal.sosfiltfilt(sos, proxy)
        r = np.array([abs(np.corrcoef(s, proxy)[0, 1])
            for s in src])
        scores = r
        z = (r - r.mean()) / (r.std() + 1e-12)
        inds = list(np.where((z > 3.0) | (r > 0.5))[0])
        report.ica_method_used = "correlation fallback"

    inds = [int(i) for i in inds]
    if scores is not None and len(inds) > ICA_MAX_BAD:
        inds = sorted(inds, key=lambda i: -
            scores[i][:ICA_MAX_BAD])
        return sorted(inds), scores

def _snap(raw, ch, store, name):
    if store is None:
        return
    x = raw.get_data(picks=[ch])[0]
    store[name] = (np.arange(len(x)) /
        raw.info["sfreq"], x)

def preprocess_subject(subject, dataset=DS_REST,
    make_epochs=True,
    snapshot_channel=None):
    rep = PreprocReport(subject=subject)
    stages = {} if snapshot_channel else None

    raw = load_raw(subject, dataset)
    rep.n_channels = len(raw.ch_names)
    rep.dur_raw_s = raw.n_times / raw.info["sfreq"]
    _snap(raw, snapshot_channel, stages, "1_raw")

    apply_own_filters(raw)
    _snap(raw, snapshot_channel, stages, "2_filtered")

    raw.crop(tmin=CROP_EDGE_S, tmax=rep.dur_raw_s -
        CROP_EDGE_S)
    rep.dur_cropped_s = raw.n_times / raw.info["sfreq"]

    raw.resample(TARGET_SFREQ, verbose="ERROR")
    rep.sfreq_out = raw.info["sfreq"]

    bads = BAD_CHANNELS.get(subject, [])
    if bads:
        raw.info["bads"] = list(bads)
        raw.interpolate_bads(reset_bads=True,
            verbose="ERROR")
        rep.interpolated = list(bads)
        print(f"{subject}: interpolated {bads}")

    raw.set_eeg_reference("average", projection=False,
        verbose="ERROR")
    _snap(raw, snapshot_channel, stages, "3_referenced")

    raw_fit = raw.copy().filter(1_freq=ICA_HIGHPASS,
        h_freq=None,
        verbose="ERROR")

    ica = mne.preprocessing.ICA(
        n_components=ICA_N_COMPONENTS,
        method=ICA_METHOD,
        fit_params=dict(extended=True) if ICA_METHOD ==
            "infomax" else None,
        max_iter=500,
        random_state=ICA_RANDOM_STATE,

```

```

)
ica.fit(raw_fit, decim=ICA_DECIM, verbose="ERROR")

bad, _ = detect_blink_components(ica, raw_fit, rep)
ica.exclude = bad
rep.ica_excluded = bad
ica.apply(raw, verbose="ERROR")
_snap(raw, snapshot_channel, stages,
    "4_ica_cleaned")

epochs = None
if make_epochs:
    epochs = mne.make_fixed_length_epochs(
        raw, duration=EPOCH_LEN_S,
        overlap=EPOCH_OVERLAP_S,
        preload=True, verbose="ERROR")
    rep.n_epochs_made = len(epochs)
    epochs.drop_bad(reject=dict(eeg=REJECT_PTP),
        verbose="ERROR")
    rep.n_epochs_kept = len(epochs)

return raw, epochs, ica, rep, stages

```

Listing 4. features.py: the 94 features

```

import numpy as np

import spectral as S

FMIN, FMAX = 0.5, 45.0
BANDS = {"delta": (0.5, 4), "theta": (4, 8), "alpha":
    (8, 13),
    "beta": (13, 25), "gamma": (25, 45)}

REGIONS = {
    "global": ["Fp1", "Fp2", "F7", "F3", "Fz", "F4",
        "F8", "T7", "C3", "Cz",
        "C4", "T8", "P7", "P3", "Pz", "P4",
        "P8", "O1", "O2"],
    "frontal": ["Fp1", "Fp2", "F7", "F3", "Fz", "F4",
        "F8"],
    "central": ["C3", "Cz", "C4"],
    "temporal": ["T7", "T8", "P7", "P8"],
    "parietal": ["P3", "Pz", "P4"],
    "occipital": ["O1", "O2"],
    "posterior": ["P3", "P4", "Pz", "O1", "O2", "P7",
        "P8"],
}

FIT_RANGE = (2.0, 40.0)
PEAK_WINDOW = (6.0, 14.0)
PEAK_THRESHOLD = 0.1

def relative_band_powers(freqs, psd):
    total = S.band_power(freqs, psd, FMIN, FMAX)
    return {b: S.band_power(freqs, psd, lo, hi) / total
        for b, (lo, hi) in BANDS.items()}

def spectral_entropy(freqs, psd):
    m = (freqs >= FMIN) & (freqs <= FMAX)
    p = psd[:, m] / psd[:, m].sum(axis=1, keepdims=True)
    h = -(p * np.log(p + 1e-300)).sum(axis=1)
    return h / np.log(m.sum())

def spectral_edge(freqs, psd, q):
    m = (freqs >= FMIN) & (freqs <= FMAX)
    f = freqs[m]
    out = np.empty(psd.shape[0])
    for c in range(psd.shape[0]):
        cum = np.cumsum(psd[c, m])
        out[c] = np.interp(q, cum / cum[-1], f)
    return out

def rms_frequency(freqs, psd, fs):
    gain2 = (2 * np.sin(np.pi * freqs / fs)) ** 2
    ratio = (psd * gain2).sum(axis=1) / psd.sum(axis=1)
    return fs / (2 * np.pi) * np.sqrt(ratio)

def hjorth(data, fs):
    d1 = np.diff(data, axis=-1)
    d2 = np.diff(d1, axis=-1)
    v0 = data.var(axis=-1).mean(axis=0)
    v1 = d1.var(axis=-1).mean(axis=0)
    v2 = d2.var(axis=-1).mean(axis=0)
    mobility = np.sqrt(v1 / v0)
    complexity = np.sqrt(v2 / v1) / mobility
    return {"activity": v0,
        "mobility_hz": mobility * fs / (2 * np.pi),
        "complexity": complexity}

def alpha_peak(freqs, spectrum):

```

```

    lf, lp = np.log10(freqs[1:]), np.log10(spectrum[1:])
    f = freqs[1:]
    fit_m = (f >= FIT_RANGE[0]) & (f <= FIT_RANGE[1]) &
~((f >= PEAK_WINDOW[0]) & (f <= PEAK_WINDOW[1]))
    slope, intercept = np.polyfit(lf[fit_m], lp[fit_m],
1)
    resid = lp - (intercept + slope * lf)

    win = np.where((f >= PEAK_WINDOW[0]) & (f <=
PEAK_WINDOW[1]))[0]
    k = win[np.argmax(resid[win])]
    height = float(resid[k])
    on_edge = k in (win[0], win[-1])
    present = (not on_edge) and height >= PEAK_THRESHOLD

    iaf = np.nan
    if present:
        y0, y1, y2 = resid[k - 1], resid[k], resid[k +
1]
        denom = y0 - 2 * y1 + y2
        shift = 0.5 * (y0 - y2) / denom if denom != 0
    else 0.0
    iaf = float(f[k] + shift * (f[1] - f[0]))

    return {"present": int(present), "iaf": iaf,
"height": height,
"slope": float(slope), "intercept":
float(intercept)}

def channel_features(freqs, psd, epochs_data, fs):
    rel = relative_band_powers(freqs, psd)
    hj = hjorth(epochs_data, fs)
    feats = {"rel_{b}": rel[b] for b in BANDS}
    feats["logTAR"] = np.log10(rel["theta"] /
rel["alpha"])
    feats["logSlowFast"] = np.log10((rel["delta"] +
rel["theta"]) /
                                (rel["alpha"] +
rel["beta"]))
    feats["entropy"] = spectral_entropy(freqs, psd)
    feats["SEF50"] = spectral_edge(freqs, psd, 0.50)
    feats["SEF95"] = spectral_edge(freqs, psd, 0.95)
    feats["hjorth_logActivity"] =
np.log10(hj["activity"])
    feats["hjorth_mobility"] = hj["mobility_hz"]
    feats["hjorth_complexity"] = hj["complexity"]
    return feats

def region_average(values, ch_names, region):
    idx = [ch_names.index(c) for c in REGIONS[region]]
    return float(np.mean(values[idx]))

```

Listing 5. ml.py: models, nested cross-validation and statistics

```

import sys
import warnings
from pathlib import Path

import numpy as np
import pandas as pd
from scipy.stats import rankdata
from sklearn.base import BaseEstimator, TransformerMixin
from sklearn.dummy import DummyClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.feature_selection import SelectKBest,
f_classif
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, roc_curve
from sklearn.model_selection import GridSearchCV,
RepeatedStratifiedKFold, StratifiedKFold
from sklearn.neighbors import KNeighborsClassifier
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC

sys.path.append(str(Path(__file__).resolve().parent))
from eegio import PROCESSED

TASKS = {
    "AD_vs_CN": ["CN", "AD"],
    "FTD_vs_CN": ["CN", "FTD"],
    "AD_vs_FTD": ["FTD", "AD"],
    "3class": ["CN", "FTD", "AD"],
}
FEATURE_SETS = ["region", "channel"]
MODELS = ["dummy", "logreg", "linsvm", "rbfsvm", "rf",
"knn"]

```

```

MODEL_NAMES = {"dummy": "Chance", "logreg": "Logistic
regression",
"linsvm": "Linear SVM", "rbfsvm": "RBF
SVM", "rf": "Random forest",
"knn": "k-nearest neighbours"}
PRIMARY = ("AD_vs_CN", "region", "logreg")
META = ["participant_id", "group", "age", "sex",
"n_epochs"]
N_FOLDS, N_REPEATS, INNER_FOLDS = 5, 10, 5
N_BOOT = 2000
SEED = 2026

def load_task(task, feature_set):
    classes = TASKS[task]
    source = "region" if feature_set == "age_sex" else
feature_set
    df = pd.read_csv(PROCESSED /
f"features_{source}.csv")
    df =
df[df.group.isin(classes)].reset_index(drop=True)
    if feature_set == "age_sex":
        X = pd.DataFrame({"age": df.age.astype(float),
"sex_male":
(df.sex.astype(str).str.upper() ==
"M").astype(float)})
    else:
        X = df.drop(columns=[c for c in META if c in
df.columns]).astype(float)
        y = df.group.map({g: i for i, g in
enumerate(classes)}).to_numpy()
        return X.to_numpy(), y,
df.participant_id.to_numpy(), list(X.columns),
classes

class ConfoundRegressor(BaseEstimator,
TransformerMixin):
    def __init__(self, n_conf=2):
        self.n_conf = n_conf

    def _split(self, X):
        X = np.asarray(X, dtype=float)
        C = np.column_stack([np.ones(len(X)), X[:, -
self.n_conf:]])
        return X[:, :-self.n_conf], C

    def fit(self, X, y=None):
        F, C = self._split(X)
        self.beta_ = np.linalg.lstsq(C, F,
rcond=None)[0]
        return self

    def transform(self, X):
        F, C = self._split(X)
        return F - C @ self.beta_

def model_spec(name, n_features, n_confounds=0):
    if name == "dummy":
        return Pipeline([("clf",
DummyClassifier(strategy="constant", constant=0))]),
None
    k_grid = [k for k in (10, 25, 50, 100) if k <
n_features] + ["all"]
    clf, grid = {
        "logreg":
(LogisticRegression(class_weight="balanced",
max_iter=5000),
{"clf__C": [0.001, 0.01, 0.1, 1, 10,
100]}),
        "linsvm": (SVC(kernel="linear",
class_weight="balanced", probability=True,
random_state=0),
{"clf__C": [0.001, 0.01, 0.1, 1,
10]}),
        "rbfsvm": (SVC(kernel="rbf",
class_weight="balanced", probability=True,
random_state=0),
{"clf__C": [0.1, 1, 10, 100],
"clf__gamma": ["scale"]}),
        "rf": (RandomForestClassifier(n_estimators=200,
class_weight="balanced", random_state=0),
{"clf__max_features": ["sqrt", 0.3],
"clf__min_samples_leaf": [1, 3]}),
        "knn": (KNeighborsClassifier(),
{"clf__n_neighbors": [3, 5, 7, 9, 15]}),
    ][name]
    if name == "rf":
        k_grid = [k for k in k_grid if k == "all" or k
== 25]

```

```

        steps = [("impute", SimpleImputer(strategy="median",
keep_empty_features=True))]
        if n_confounds:
            steps.append(("deconfound",
ConfoundRegressor(n_conf=n_confounds)))
        steps += [("scale", StandardScaler()), ("select",
SelectKBest(f_classif)), ("clf", clf)]
        pipe = Pipeline(steps)
        return pipe, {"**grid", "select_k": k_grid}

def run_cv(X, y, name, n_repeats=N_REPEATS, seed=SEED,
n_jobs=-1, inner_folds=INNER_FOLDS,
n_confounds=0):
    warnings.filterwarnings("ignore")
    n_classes = len(np.unique(y))
    outer = RepeatedStratifiedKFold(n_splits=N_FOLDS,
n_repeats=n_repeats, random_state=seed)
    proba = np.full((n_repeats, len(y), n_classes),
np.nan)
    pred = np.full((n_repeats, len(y)), -1)
    n_feat = X.shape[1] - n_confounds
    sel = np.zeros(n_feat)
    coef = np.zeros(n_feat)
    params = []
    scoring = "roc_auc" if n_classes == 2 else
"balanced_accuracy"
    for f, (tr, te) in enumerate(outer.split(X, y)):
        pipe, grid = model_spec(name, n_feat,
n_confounds)
        if grid is None:
            est = pipe.fit(X[tr], y[tr])
        else:
            inner = StratifiedKFold(inner_folds,
shuffle=True, random_state=seed + f)
            gs = GridSearchCV(pipe, grid,
scoring=scoring, cv=inner, n_jobs=n_jobs)
            est = gs.fit(X[tr], y[tr]).best_estimator_
            params.append(gs.best_params_)
            mask =
est.named_steps["select"].get_support()
            sel += mask
            c = getattr(est.named_steps["clf"], "coef_",
None)
            if c is not None and n_classes == 2 and not
hasattr(c, "toarray"):
                coef[mask] += np.ravel(c)
            proba[f // N_FOLDS, te] =
est.predict_proba(X[te])
            pred[f // N_FOLDS, te] = est.predict(X[te])
        n_outer = N_FOLDS * n_repeats
        return {"proba": proba, "pred": pred, "params":
params,
            "sel_freq": sel / n_outer, "coef": coef /
n_outer}

def _auc_binary(y, s):
    r = rankdata(s, axis=1)
    pos = y == 1
    n1, n0 = pos.sum(), (~pos).sum()
    return (r[:, pos].sum(axis=1) - n1 * (n1 + 1) / 2) /
(n1 * n0)

def _scores(y, proba, pred):
    k = proba.shape[2]
    recall = np.stack([(pred[:, y == c] ==
c).mean(axis=1) for c in range(k)], axis=1)
    out = {"bacc": recall.mean(axis=1)}
    if k == 2:
        out["auc"] = _auc_binary(y, proba[:, :, 1])
        out["sens"], out["spec"] = recall[:, 1],
recall[:, 0]
    else:
        out["auc"] = np.mean([_auc_binary((y ==
c).astype(int), proba[:, :, c])
            for c in range(k)],
axis=0)
    return out

def summarise(y, proba, pred, n_boot=N_BOOT, seed=0):
    s = _scores(y, proba, pred)
    point = {k: float(v.mean()) for k, v in s.items()}
    point["auc_sd_repeats"] = float(s["auc"].std())
    rng = np.random.default_rng(seed)
    idx_by_class = [np.where(y == c)[0] for c in
np.unique(y)]
    boot = {"auc": np.empty(n_boot), "bacc":
np.empty(n_boot)}
    for i in range(n_boot):
        b = np.concatenate([rng.choice(ix, len(ix),
replace=True) for ix in idx_by_class])
        sb = _scores(y[b], proba[:, b], pred[:, b])
        for k in boot:
            boot[k][i] = sb[k].mean()
        for k in boot:
            point[f"{k}_lo"], point[f"{k}_hi"] = (float(v)
for v in np.percentile(boot[k], [2.5, 97.5]))
        return point

def mean_confusion(y, pred, n_classes):
    cms = [confusion_matrix(y, p,
labels=range(n_classes), normalize="true") for p in
pred]
    return np.mean(cms, axis=0)

def main_score(y, proba):
    if proba.shape[2] == 2:
        return float(_auc_binary(y, proba[:, :,
1])).mean()
    return float(np.mean([_auc_binary((y ==
c).astype(int), proba[:, :, c]).mean()
        for c in
range(proba.shape[2])]))

def mean_roc(y, proba, grid=np.linspace(0, 1, 101)):
    tprs = []
    for p in proba:
        fpr, tpr, _ = roc_curve(y, p[:, 1])
        tprs.append(np.interp(grid, fpr, tpr))
    tpr = np.mean(tprs, axis=0)
    tpr[0] = 0.0
    return grid, tpr

def auc_per_repeat(y, proba):
    if proba.shape[2] == 2:
        return _auc_binary(y, proba[:, :, 1])
    return np.mean([_auc_binary((y == c).astype(int),
proba[:, :, c])
        for c in range(proba.shape[2])]),
axis=0)

def paired_difference(y, proba_a, proba_b,
n_boot=N_BOOT, seed=0):
    d0 = float(auc_per_repeat(y, proba_b).mean() -
auc_per_repeat(y, proba_a).mean())
    rng = np.random.default_rng(seed)
    idx_by_class = [np.where(y == c)[0] for c in
np.unique(y)]
    boot = np.empty(n_boot)
    for i in range(n_boot):
        b = np.concatenate([rng.choice(ix, len(ix),
replace=True) for ix in idx_by_class])
        boot[i] = (auc_per_repeat(y[b], proba_b[:,
b]).mean()
            - auc_per_repeat(y[b], proba_a[:,
b]).mean())
    lo, hi = np.percentile(boot, [2.5, 97.5])
    p = min(1.0, 2 * min(np.mean(boot <= 0),
np.mean(boot >= 0)))
    return d0, float(lo), float(hi), float(p)

def outer_splits(y, n_repeats=N_REPEATS, seed=SEED):
    outer = RepeatedStratifiedKFold(n_splits=N_FOLDS,
n_repeats=n_repeats, random_state=seed)
    return [(f // N_FOLDS, te) for f, (_, te) in
enumerate(outer.split(np.zeros((len(y), 1)), y))]

def foldwise_auc(y, proba, seed=SEED):
    vals = []
    for r, te in outer_splits(y, proba.shape[0], seed):
        vals.append(float(auc_per_repeat(y[te],
proba[r:r + 1, te])[0]))
    return np.array(vals)

```

4 Implementation

4.1 Description

The design was implemented as a Python 3 software pipeline on an ordinary Windows laptop, using NumPy and SciPy for signal processing, MNE-Python [16] for EEG input/output, montages, resampling and ICA, scikit-learn [17] for the classifiers and cross-validation, pandas for tables and matplotlib for figures. The DSP itself (filter design and application, the Welch estimator, every feature and the validation protocol) is our own code; library routines were used where a tested implementation exists, and every one of them was checked against a known answer or a second implementation.

Software structure. Six reusable modules contain the methods, and numbered scripts run the analysis in order (Table 14). Each script starts with a checkpoint that must pass before its results are trusted; for example, the feature script first rebuilds earlier features and stops if they differ by more than 10^{-3} .

Table 9. *Software modules and analysis scripts*

Module or scripts	Responsibility
eegio.py	Paths, reading BIDS/EEGLAB files, channel renaming and montage, participant table
filters.py	FIR (Kaiser, Hamming) and IIR Butterworth band-pass design, 50 Hz notch, zero-phase application
preprocess.py	Crop, resample, faulty-electrode repair, CAR, ICA with automatic blink detection, epoching and rejection
spectral.py	Our Welch estimator for continuous data and for epochs, band power by trapezoid integration
features.py	The 13 measures, regional averaging, 1/f-corrected alpha-peak detection
ml.py	Model pipelines and grids, nested cross-validation, AUC, bootstrap, permutation, paired comparisons, confound regression
Scripts 00–11	Data checks, first spectra, filter design and verification, preprocessing demonstration, comparison with the authors' derivatives
Scripts 12–17	Batch preprocessing of all recordings, Welch validation, cohort spectra, feature extraction and ranking
Scripts 19–25	Eyes-open data (ds006036), alpha reactivity, EEG fingerprinting and the participant-ID check
Scripts 26–29	Validation checkpoint, the 52 classification runs, permutation test, sex check
Scripts 30–43	Feature families, region against channel, cleaning ablation, learning curve, error analysis, robustness and confounds
44_live_demo.py	Scores one person with a model trained on everyone else and plots the result

Reproducibility. Every random step has a fixed seed (ICA 97, cross-validation 2026), all splits depend only on the labels and the seed, and intermediate results are cached. Re-running the full preprocessing for the whole cohort reproduced the stored features to a maximum relative difference of 1.2×10^{-7} , and every later stage reproduced the earlier results exactly before adding anything new. Preprocessing all 88 recordings takes about 15 minutes; scoring one person, from cleaned data to a probability, takes about 6 seconds.

Demonstration tool. For the live demonstration, 44_live_demo.py takes one participant, loads their cleaned epochs, recomputes their spectrum and 94 features (and checks that these are identical to the stored feature table), trains the main pipeline with its inner tuning loop on every other AD and CN person, and outputs P(AD). As in cross-validation, the person being scored is never seen by the model that scores them. Fig. 26 shows the output for an AD patient whom the model misses: the posterior

spectrum (black) keeps a clear alpha peak at 11.5 Hz and the theta/alpha ratio lies at the low edge of the AD group, so the model scores 0.21. Such cases are analysed in Section 5.3.3.

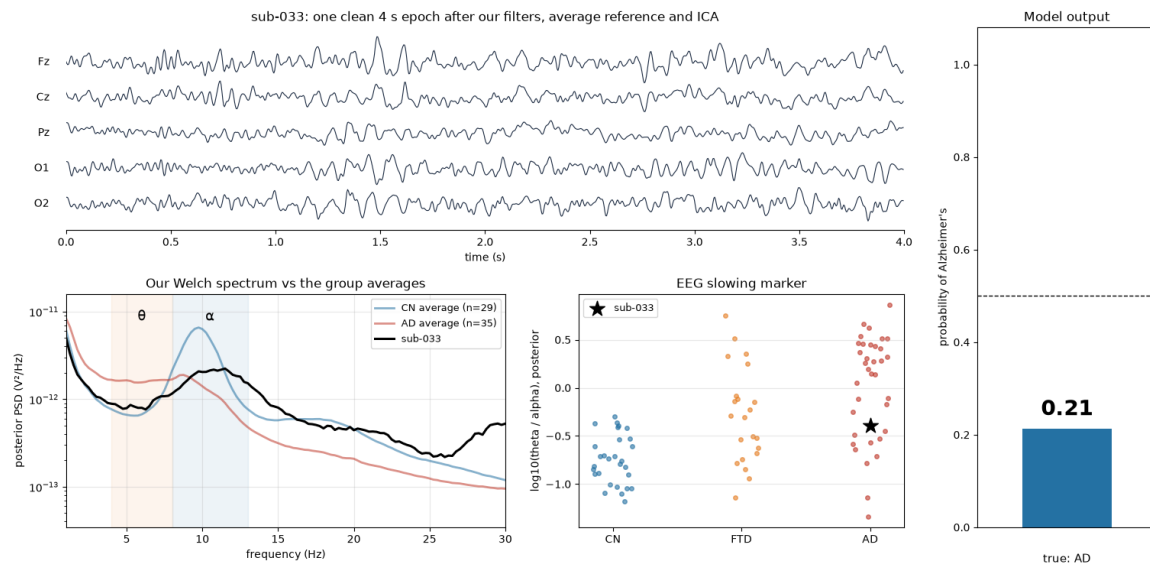


Figure 26. Output of the demonstration tool for sub-033 (AD): one clean epoch, the posterior Welch spectrum against the group averages, the theta/alpha marker, and the model's probability of AD

5 Design Analysis and Evaluation

5.1 Novelty

We do not claim a new biomarker: EEG slowing in AD is well established [3], and other groups also evaluate on people unseen during training. Our contributions are narrower and can be checked:

- **A fully verified DSP chain.** Every stage was designed to a written specification and verified: both filters meet their specification after correction, our Welch estimator equals SciPy's to machine precision, and the complete pipeline reruns to 10^{-7} .
- **A quantified contribution for each DSP step.** The cleaning ablation measures what filtering, re-referencing, rejection and ICA each add (the average reference alone adds 0.13 AUC), which published results on this dataset rarely report.
- **A result declared before it was computed.** The headline analysis, the robustness criteria and their pass levels were written down in advance, and the headline was tested against 200 label permutations, so it cannot be the best of the 52 runs chosen afterwards.
- **A flaw found in public data.** The README of ds006036 states that its participant numbers match ds004504. Comparing each person's spectral "fingerprint" showed that they mostly do not (Section 5.3.3). A 2026 preprint [26] on this cohort joined the two datasets by participant number, so the finding matters beyond this project.

5.2 Design Considerations (PO(c))

The design was shaped by its intended use: a low-cost screening aid in hospitals that already own EEG equipment. The considerations below influenced the specification, the validation and the way results are reported.

5.2.1 Considerations to public health and safety

- **Non-invasive and safe.** EEG records voltages only; it involves no radiation, contrast agent, needle or stimulation, and it is safe for elderly and frail people.
- **Screening, never diagnosis.** The output is framed as "refer for full assessment". About one patient in four is missed (sensitivity 0.73) and one healthy person in five is flagged (specificity 0.80), so a clinician must confirm every result.
- **Adjustable operating point.** Because the model outputs a probability, a clinic can lower the threshold to miss fewer patients at the cost of more referrals; the AUC describes the trade-off at every threshold.
- **Avoiding false reassurance.** A low score must not be read as "healthy" in a person with symptoms; the instructions for use state this explicitly.

5.2.2 Considerations to environment

The system adds almost no environmental burden. It runs on existing EEG machines and an ordinary computer; it needs no new hardware, imaging consumables or reagents, only the conductive gel and electrodes already used for routine EEG. Processing one recording uses seconds of computing time, and the whole cohort was processed in about 15 minutes on a laptop, a negligible amount of energy

compared with an MRI or PET examination. A low-cost few-electrode headset, the long-term goal, would further reduce material use and electronic waste.

5.2.3 Considerations to cultural and societal needs

Dementia often carries stigma and is mistaken for normal ageing, which delays care. A low-cost, harmless test could encourage earlier help-seeking, but results must be communicated with care and privacy, in the patient's language, and with the family involved where appropriate. The test does not depend on reading, writing or education, unlike questionnaire screening, which makes it fairer for people with little formal schooling. It suits Bangladesh, where imaging is concentrated in a few cities but EEG is available in most tertiary hospitals.

5.3 Investigations (PO(d))

The investigation combined one confirmatory analysis, fixed in advance, with exploratory and robustness analyses, all on the same people, in the same order and with the same cross-validation splits, so that differences between analyses are paired comparisons person by person.

5.3.1 Design of Experiment

The investigation was organised around four questions, each with a fixed protocol:

- **RQ1 (main, confirmatory).** How well does resting EEG separate AD from CN for people never seen by the model? Answered by the single pre-declared analysis (Section 3.1.3) and a permutation test.
- **RQ2 (secondary).** Can FTD be separated from CN, and AD from FTD? Answered by the same pipeline; with three comparisons, multiple testing was taken into account.
- **RQ3 (exploratory).** What drives the result? Feature families removed or used alone, region against channel features, the cleaning ablation, the learning curve and an error analysis. All are compared with the main model person by person using a paired bootstrap of the AUC difference.
- **RQ4 (robustness, criteria fixed in advance).** Would other reasonable choices change the answer? Five criteria were written down before any of these runs (Table 15).

Beside the main analysis, a grid of 52 runs was reported in full: four comparisons (AD–CN, FTD–CN, AD–FTD, three classes), three feature sets (94 regional, 250 channel-level, and an age-plus-sex baseline), and six models (a constant chance model, logistic regression, linear and RBF SVM, random forest and k-nearest neighbours).

Table 10. Robustness criteria, written before the robustness runs

#	Criterion (AD vs CN)	Pass level
1	Seed: the main seed (2026) is typical of 20 other seeds	within 0.03 of their median
2	Scoring: fold-wise AUC instead of pooled AUC	not lower than pooled – 0.05
3	Alternatives: 2 s epochs; 100 or 200 μ V rejection; no delta band	95 % CI above 0.5 and $ \Delta \leq 0.05$
4	Stress tests: θ/α ratios only (7 features); posterior electrodes only (16)	95 % CI above 0.5
5	Age and sex: EEG adds beyond them; matched and deconfounded models	$p < 0.05$ and every CI above 0.5

5.3.2 Data Collection

No new recordings were made. The primary data are the 88 eyes-closed resting recordings of OpenNeuro ds004504 [4] (CC0 licence), recorded at 500 Hz with 19 electrodes at a median of about 13 minutes per person after cropping (AD and CN) and 12 minutes (FTD). Clinical labels (AD, FTD, CN), age, sex and MMSE come from the dataset's participant table. After cleaning, a median of 367 (AD), 382 (CN) and 320.5 (FTD) clean 4 s epochs per person entered the spectral estimate. One FTD recording was excluded for data quality (Section 3.2.3), leaving 87 people: 36 AD, 29 CN and 22 FTD. For the alpha-reactivity analysis, the eyes-open photic-stimulation recordings of ds006036 [5] were used. These are mostly eyes closed with short eye-opening tests of 3–6 s, so reactivity was measured inside ds006036 by comparing clean eyes-open windows with eyes-closed time within 30 s of each test (2 s windows, 0.5 Hz resolution). After excluding flash periods, marked artefacts and calibration segments, 45 people qualified (17 AD, 22 CN, 6 FTD).

5.3.3 Results and Analysis

Main result: AD versus CN

The pre-declared analysis gave the results in Table 16. In 200 permutations of the diagnoses, with the entire pipeline rerun each time, no shuffle reached the real score (null mean 0.500, null maximum 0.722), giving $p = 0.005$ (Fig. 29). Age and sex alone, with no EEG, reached only 0.624 [0.495, 0.744].

Table 11. Main result (AD vs CN, 94 regional features, logistic regression, nested CV)

Measure	Value	Meaning
AUC	0.811 [0.717, 0.896]	a random AD patient is ranked above a random CN participant 81 % of the time
Balanced accuracy	0.762 [0.676, 0.846]	mean of sensitivity and specificity at $p = 0.5$ (chance 0.5)
Sensitivity / specificity	0.73 / 0.80	share of AD flagged / share of CN cleared
Permutation test	$p = 0.005$	0 of 200 label shuffles reached the real AUC

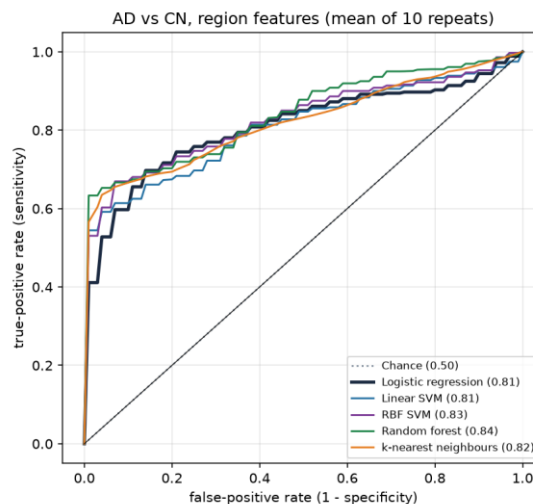


Figure 27. Cross-validated ROC curve of the main model, AD against CN

KEY RESULT

- Resting EEG separates Alzheimer's patients from healthy people with AUC 0.811 [0.717, 0.896] on people never seen by the model.
- Chance cannot explain it ($p = 0.005$), and age and sex alone reach only 0.624.

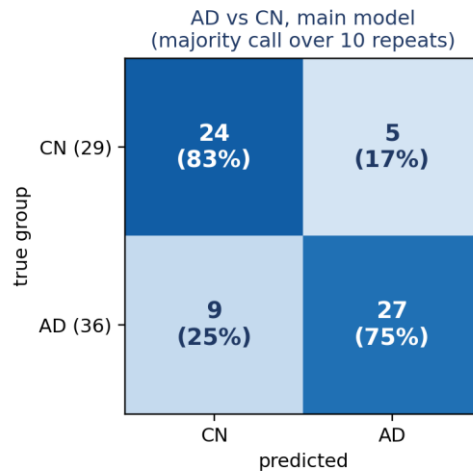


Figure 28. Confusion matrix of the main model at $p = 0.5$ (each person's majority call over the 10 repeats)

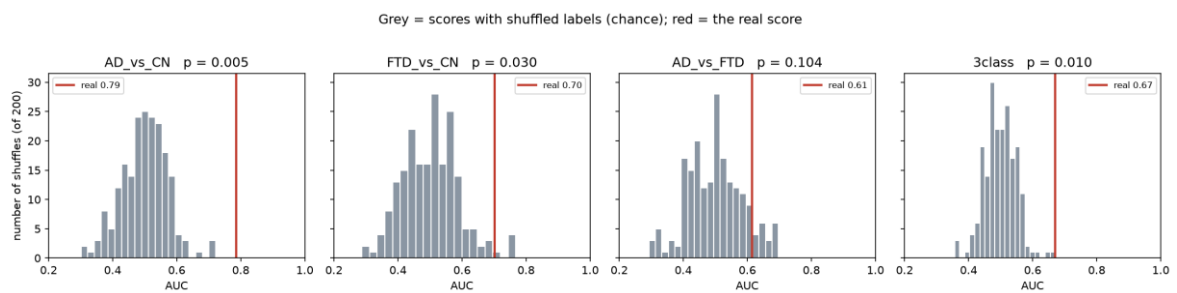


Figure 29. Permutation tests: AUC after shuffling the diagnoses (grey) against the real score (red) for each comparison

In the inner loop the model usually kept 25 features. The theta/alpha ratios of every region were selected in 96–100 % of the 50 outer folds, followed by the slow/fast ratios, the alpha-peak height and relative theta (Fig. 30): together, a description of EEG slowing. Because these features are strongly correlated, the sign of a single coefficient is not reliable; feature families are interpreted instead.

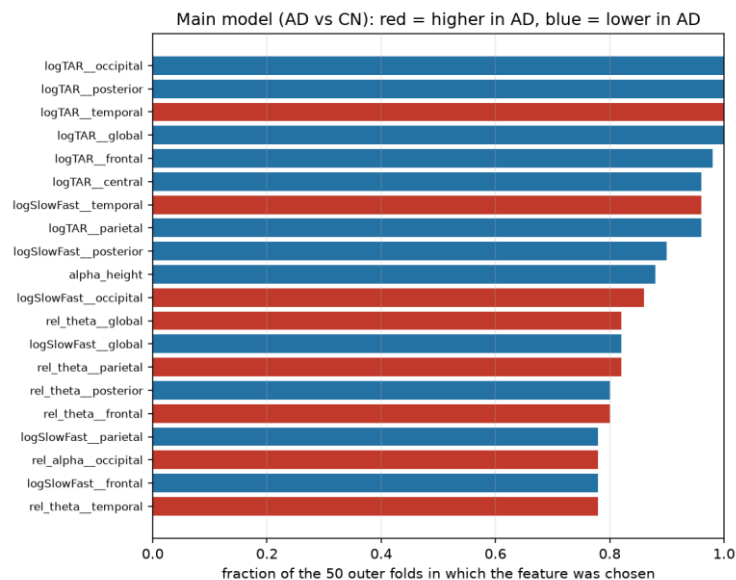


Figure 30. How often each feature was selected across the 50 outer folds (red: higher in AD, blue: lower)

All comparisons and models

Table 12. Cross-validated AUC across comparisons and models (selected rows of the 52 runs; * see text)

Comparison	Features	Logistic reg.	Linear SVM	RBF SVM	Random forest	kNN
AD vs CN	regional (94)	0.811	0.810	0.828	0.842	0.822
AD vs CN	channel (250)	0.844	0.831	0.865	0.850	0.855
FTD vs CN	regional	0.728				
FTD vs CN	channel	0.786				
AD vs FTD	regional	0.578	0.445*	0.437*		

FTD versus CN reached AUC 0.728 [0.602, 0.836] ($p = 0.030$), moderate evidence that passes a false-discovery correction but not the stricter Holm correction across the three comparisons. **AD versus FTD** was not separable: 0.578 [0.444, 0.707], $p = 0.104$. With all **three classes**, balanced accuracy was 0.532 [0.454, 0.609] against a chance level of 0.333 ($p = 0.010$). For AD versus CN, no model was clearly better than the pre-declared one: the differences (up to 0.03 for regional features) are much smaller than the uncertainty of about ± 0.09 , and the random seed alone moves the main model by 0.04 (below). Picking the best-scoring model afterwards would report the luckiest of many noisy scores. *The two SVMs scored below chance on AD vs FTD.* Our first explanation, that pooling probabilities from different fold models distorts the ranking, was tested and rejected: fold-wise AUCs were even lower (0.425 and 0.390). The real cause is the probability step of the SVMs (Platt scaling [27]), which is fitted by an internal cross-validation and can invert the ranking when there is little signal. The SVMs' own decision values gave 0.578 and 0.537, and the probability step had flipped the ranking in 38 % and 74 % of the fold models (against 4 % and 0 % for AD vs CN).

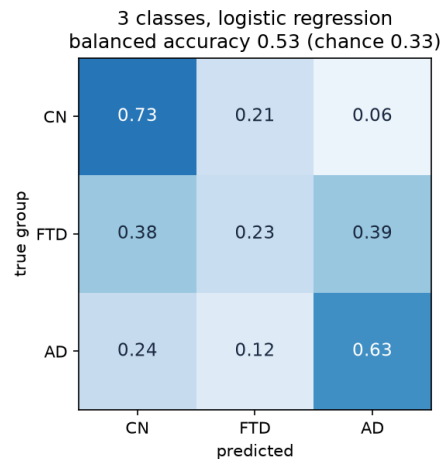


Figure 31. Three-class confusion matrix (rows sum to 1): FTD is confused with both other groups

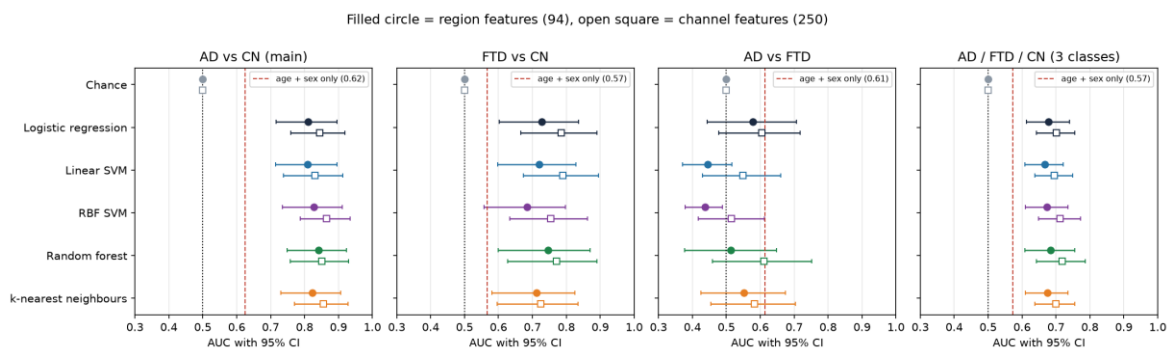


Figure 32. AUC with 95 % intervals for every model and feature set in each comparison

What drives the result

Feature families. Each family was used alone and removed in turn (Fig. 33). The information is almost entirely redundant: removing any one family changed the AD vs CN AUC by at most 0.03, the 14 ratio features alone matched all 94 (0.811), and the three alpha-peak features alone reached 0.809. Spectral-shape (0.738) and Hjorth (0.734) features alone were weaker. The result is carried by one phenomenon, EEG slowing, which every family measures from a different angle.

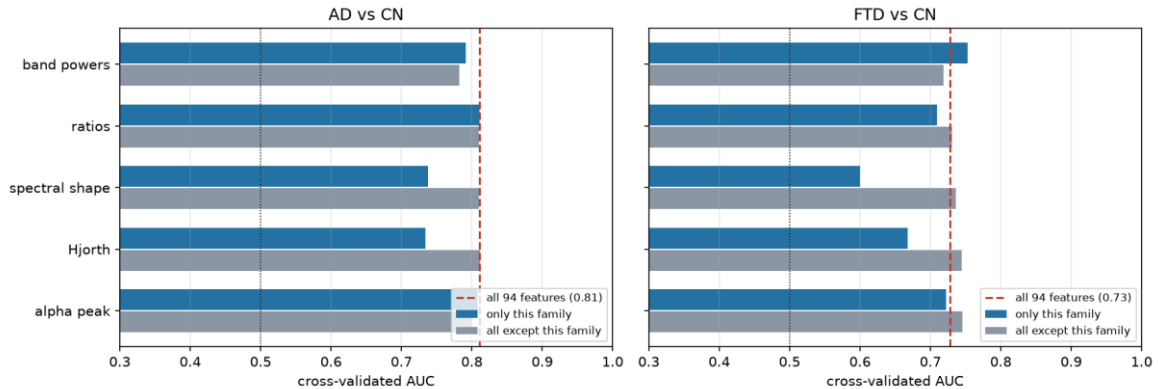


Figure 33. Each feature family alone (blue) and all except it (grey); dashed line: all 94 features

Regions against channels. Channel-level features scored higher than regional ones in all 20 paired comparisons, but for the main AD vs CN model the gain was small and not significant (+0.034 [−0.008, +0.077], $p = 0.127$): AD slowing is diffuse and survives averaging. For FTD vs CN the gain was significant (+0.058, $p = 0.008$), consistent with a more focal, frontotemporal change.

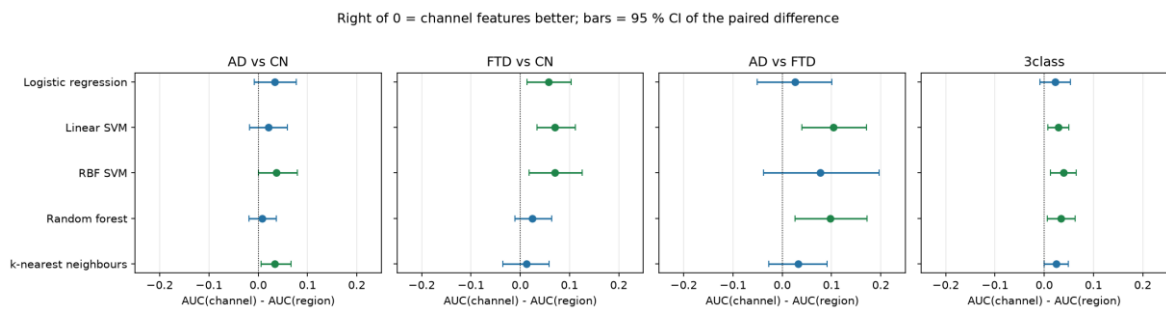


Figure 34. Channel-level minus regional AUC for every model and comparison, with 95 % paired-bootstrap intervals

Alpha reactivity. In healthy people, alpha power fell by a factor of 8.9 on average when the eyes opened (in 95.5 % of people), against only 1.7 in AD (82.4 %). All four reactivity features separated AD from CN inside ds006036 (AUC 0.743–0.789, all passing a false-discovery correction; Fig. 36). The difference came entirely from eyes-closed alpha (AUC 0.82, $p = 7 \times 10^{-4}$), while eyes-open alpha did not differ (0.54). Adding reactivity to the resting features did not improve the model (+0.014, $p = 0.43$, 32 people), because the resting features already contain eyes-closed alpha.

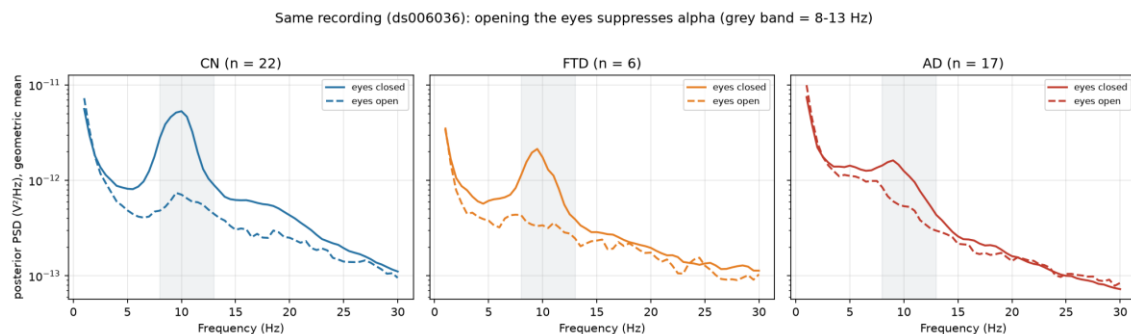


Figure 35. Group spectra with eyes closed and eyes open (ds006036): opening the eyes suppresses alpha strongly in CN but only weakly in AD

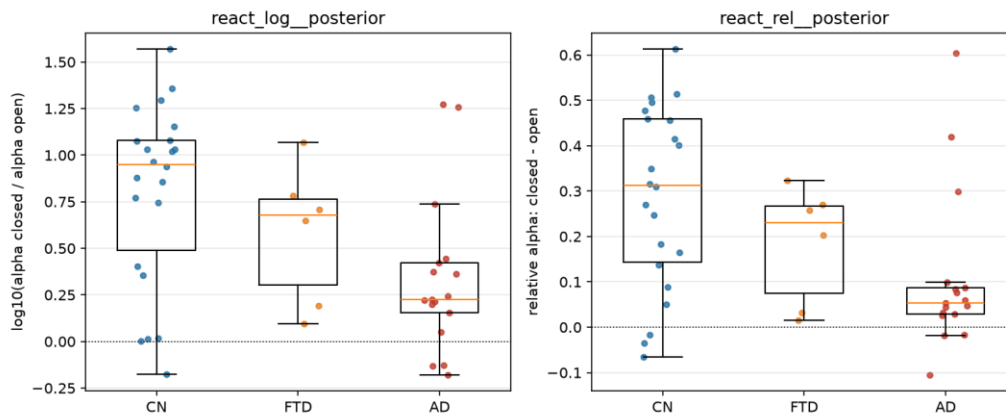


Figure 36. Alpha reactivity (eyes-closed relative to eyes-open alpha, posterior) by group

Amount of data. The learning curve is flat: the AUC rose only from 0.814 with 20 training people to 0.830 with 52 (Fig. 37). More data of the same kind would mainly narrow the confidence interval; the performance is limited by what these features contain, not by the sample size.

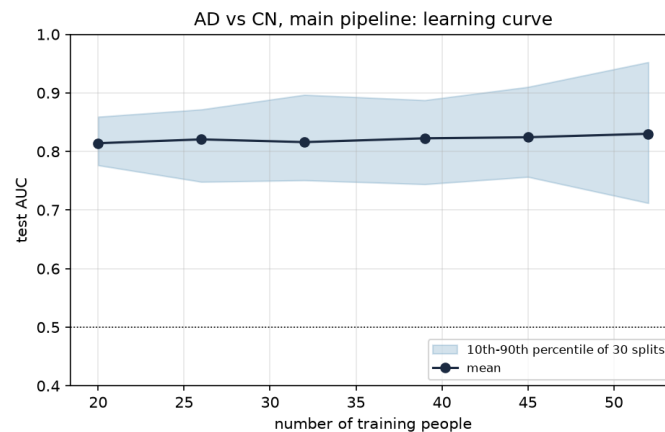


Figure 37. Test AUC against the number of training people (10th–90th percentile band of 30 splits)

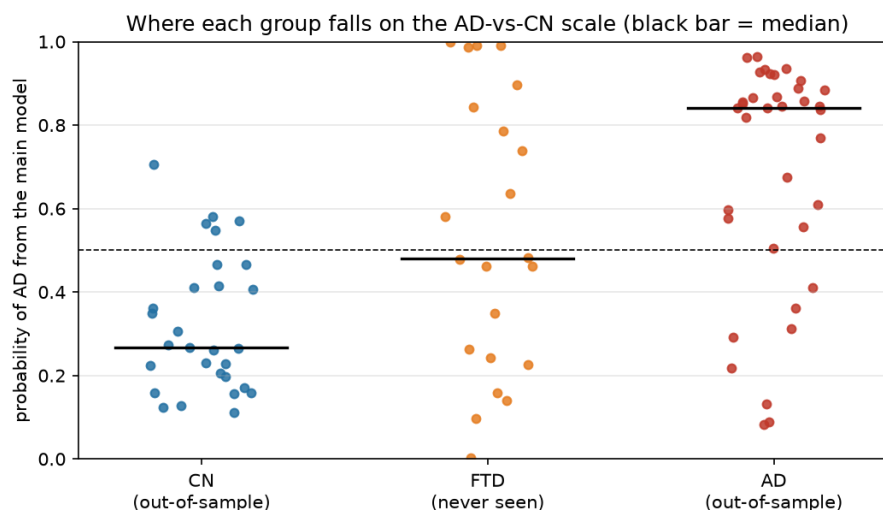


Figure 38. Out-of-sample probability of AD for every person; FTD patients (never used for training) fall between CN and AD

Errors and FTD. Nine of 36 AD patients were missed and 5 of 29 CN participants were wrongly flagged (wrong in more than half of the 10 repeats). The errors did not differ from correct cases in MMSE, age, data loss, number of ICA components removed or recording length (all $p > 0.18$), and

missed patients were not milder (median MMSE 18 against 20). The false alarms were all borderline (0.55–0.71), but some misses were confident (0.08 and 0.09): some AD patients simply have a normal-looking EEG. FTD patients, scored by the AD vs CN model that never saw any FTD data, fell between the groups (medians CN 0.27, FTD 0.48, AD 0.84; Fig. 38). FTD shares the slowing, which explains why AD and FTD cannot be separated with these features.

Robustness

All five pre-declared criteria were met (Table 18, Fig. 39).

Table 13. *Robustness results*

Check	AD vs CN AUC	Criterion
20 other seeds	median 0.807 (range 0.780–0.821); seed 2026 ranks 8 of 21	1 · met
Fold-wise instead of pooled AUC	0.826 (no fold below 0.5)	2 · met
2 s epochs / 100 μ V / 200 μ V / no delta	0.812 / 0.793 / 0.805 / 0.806 (none differs significantly)	3 · met
θ/α ratios only (7) / posterior only (16)	0.827 [0.729, 0.910] / 0.828 [0.731, 0.914]	4 · met
Age and sex	EEG adds strongly ($p = 8.7 \times 10^{-7}$); matched and deconfounded 0.753–0.789	5 · met

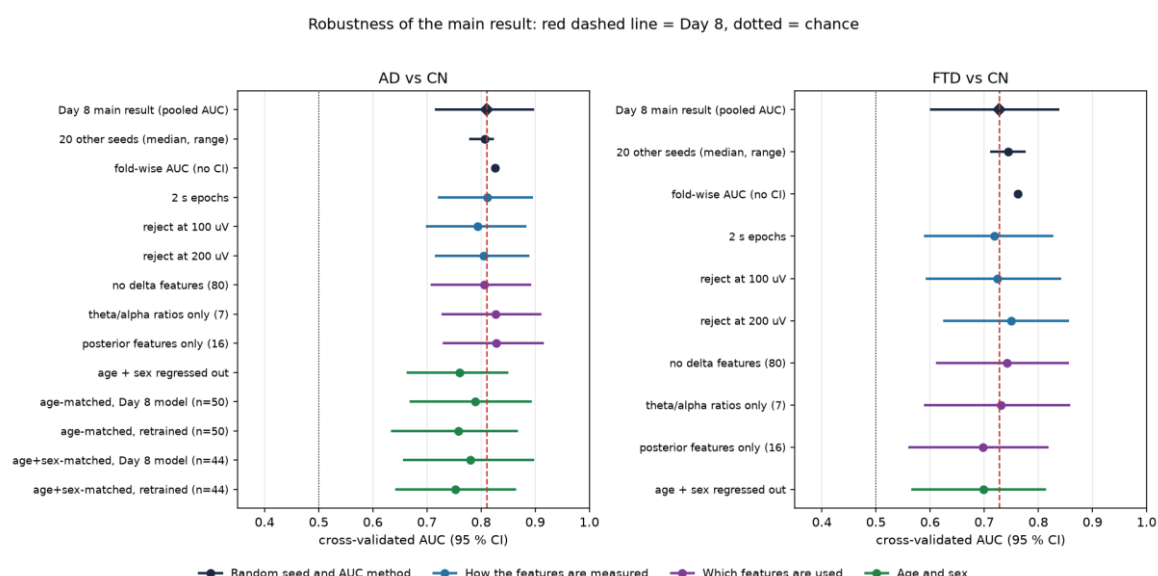


Figure 39. *Every alternative analysis with its 95 % interval; dashed line: main result, dotted line: chance*

The seed results show that 2026 was not a lucky seed (Fig. 41). Pooling probabilities across fold models costs about 0.015, so the headline is slightly conservative. A stricter 100 μ V limit removed more data from both groups (AD 5.3 $\rightarrow$ 12.9 %, CN 5.6 $\rightarrow$ 11.3 %) without a group difference ($p = 0.43$), so it did not selectively remove AD slow waves. Posterior electrodes alone, far from the eyes, worked as well as all electrodes, ruling out eye artefacts as the source of the result.

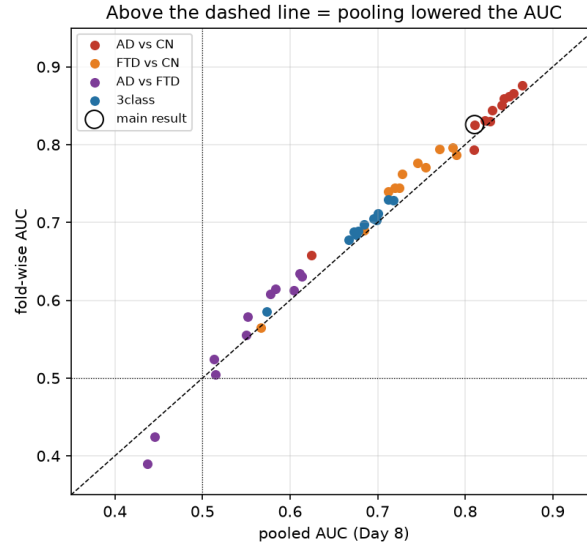


Figure 40. Fold-wise against pooled AUC for all runs; points above the diagonal are runs in which pooling lowered the AUC

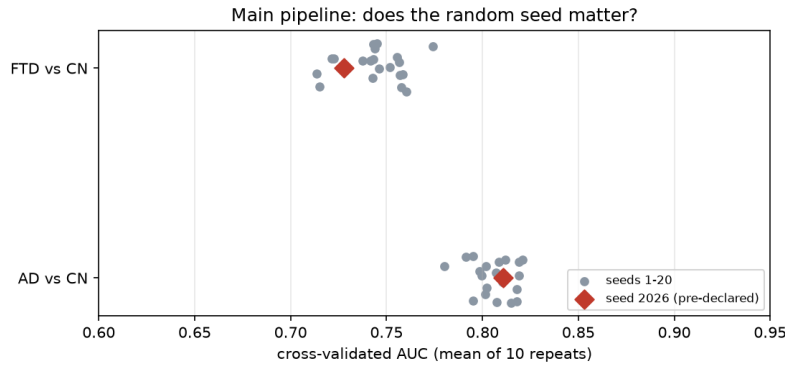


Figure 41. Main pipeline with 20 other random seeds; the pre-declared seed (red) is typical

Age and sex. Because the AD group contains more women, sex was controlled in three ways. (i) In 22 pairs matched for sex and age (within 3 years), where neither can help, the main model scored 0.781 [0.658, 0.896] (0.753 when retrained on the pairs only). (ii) In a logistic model with age and sex, adding the out-of-sample EEG score improved the fit strongly (likelihood-ratio $\chi^2(1) = 2(\ell_1 - \ell_0) = 24.2$, $p = 8.7 \times 10^{-7}$): at equal age and sex, each standard deviation of the EEG score multiplies the odds of AD by 5.7 [2.35, 13.81]. (iii) Regressing age and sex out of every feature inside each training fold [9], using only the training part to estimate the regression,

$$\tilde{\mathbf{Z}} = \mathbf{Z} - \mathbf{C}\hat{\boldsymbol{\beta}}, \quad \hat{\boldsymbol{\beta}} = (\mathbf{C}_{\text{tr}}^T \mathbf{C}_{\text{tr}})^{-1} \mathbf{C}_{\text{tr}}^T \mathbf{Z}_{\text{tr}} \quad (18)$$

(with $\mathbf{C} = [1, \text{age}, \text{sex}]$) lowered the AUC to 0.761 [0.665, 0.848], the only significant drop of all checks (-0.050 , $p = 0.012$). Because sex and diagnosis are correlated in this sample, this removes some genuine disease signal as well; the matched and within-sex results (women 0.879, men 0.796) suggest over-correction. Healthy women were not pushed towards AD (median probability 0.26 against 0.33 for healthy men). Controlled three ways, the AUC stays at 0.75–0.79.

ROBUSTNESS VERDICT

- All five criteria written down before the runs were met.
- No reasonable alternative moved the AUC by more than 0.02; controlling age and sex three ways gave 0.75–0.79.
- The result does not depend on the random seed, the scoring method, the epoch length, the rejection limit, the delta band or eye artefacts.

A problem in the public data

While linking ds006036 to ds004504 for alpha reactivity, the same-numbered recordings disagreed completely: across the two datasets, total power, alpha frequency and relative alpha were uncorrelated (ρ between -0.01 and 0.02). We therefore compared every pair of recordings by their spectral fingerprint (the z-scored log-spectrum, 2–30 Hz, 19 channels; Fig. 42). Within one dataset, eyes-open and eyes-closed data of the same person matched in 67 % of cases (top-1). Across datasets, the same participant number matched in only 3 %, which is chance. Instead, 53 of 66 ds006036 recordings had a mutual best match in ds004504, 52 of them within the same diagnostic group, and only 2 under the same number. The matches follow long runs of constant number shifts (for example, AD 001–021 shifted by +11, CN 038–055 by +6, FTD 067–076 by +6), which a coincidence cannot produce. Group labels agree, so all group-level results are safe, but per-person age or MMSE links between the datasets are unreliable, and cross-dataset joins must use the EEG pairing.

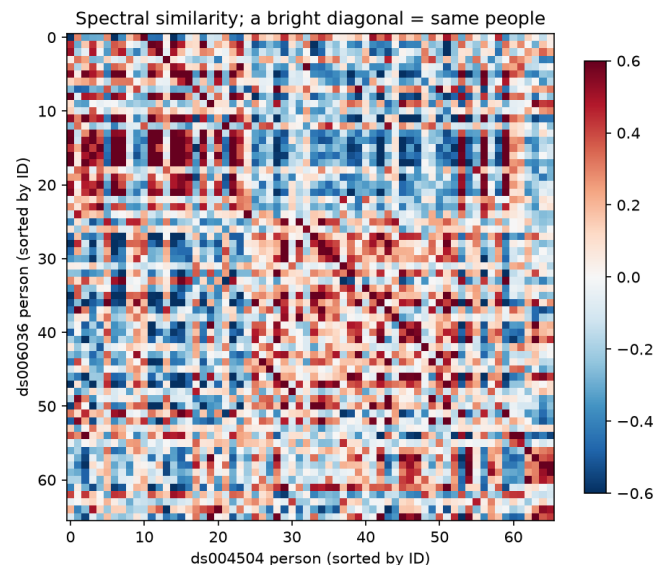


Figure 42. *Spectral similarity between recordings of the two datasets sorted by participant number; a matching numbering would show a bright diagonal*

5.3.4 Interpretation and Conclusions on Data

- **Resting EEG separates AD from healthy people well and reliably.** The pre-declared AUC of 0.811 [0.717, 0.896] comes from people never seen by the model, is far beyond chance ($p = 0.005$), and holds under every reasonable alternative and under three controls for age and sex (0.75–0.83).
- **The signal is EEG slowing, and it is simple.** Seven theta/alpha ratios carry it all (0.827), it is present at the back of the head (0.828), and every feature family measures the same thing. This matches the known physiology and makes the model explainable.
- **DSP design decides the answer.** The average reference is worth 0.13 AUC, and artefact removal changes what the score means: without ICA a higher number is partly produced by eye movements.
- **The approach screens but does not differentiate.** FTD versus CN is moderate (0.728), and AD versus FTD fails (0.578) because both dementias slow the EEG. Separating them would need information about where changes occur, not only how slow the EEG is.

- **About one patient in four has a normal-looking EEG.** Errors are not explained by severity or data quality, so the realistic role is a low-cost first step that decides who is referred for full assessment.
- **Honest methods change the numbers that are reported.** Feature selection on all data turned pure noise into AUC 0.78, and choosing the best model afterwards would have reported 0.865 instead of 0.811.

5.4 Limitations of Tools (PO(e))

Table 14. *Limitations of the tools used and how they were handled*

Tool or resource	Limitation and its effect	How we handled it
Dataset	65 AD and CN people from one hospital; wide CI (0.72–0.90); no external test set	CIs, permutation test and robustness checks reported; external validation planned
Recording hardware	19 electrodes in the 10–20 layout, ear reference: coarse spatial detail, no source localisation	CAR; regional features; posterior-only check
Spectral estimation	Resolution 1/T against variance 1/K: 4 s segments limit resolution to 0.25 Hz	2 s epochs tested: same AUC (0.812)
ICA	Iterative and random; no dedicated eye electrode, so blinks found via Fp1/Fp2	Fixed seed 97; one component per person; equal data loss checked
IIR filtering	Non-linear phase in a single pass; very long FIR alternative	Zero-phase forward–backward filtering, design at half the specification
SVM probabilities	Platt scaling can invert the ranking when the signal is weak	Decision values checked; logistic regression as the main model
Metadata	Participant numbers differ between ds004504 and ds006036	EEG-based pairing; no per-person age or MMSE claims
Computing	Nested CV with permutations is costly (52 runs, 200 reruns)	Permutation test with 2 outer repeats per shuffle; caching

5.5 Impact Assessment (PO(f))

The impact of the system depends on how it is used. The assessment below assumes use as a screening aid under clinical supervision.

5.5.1 Assessment of Societal and Cultural Issues

A test based on routine EEG could bring dementia screening to district and divisional hospitals that have EEG machines but no PET or specialist memory clinic, reducing the travel and cost burden on families. Earlier recognition lets families plan care and finances. The risk is stigma and anxiety, so results must be framed as "further assessment recommended", never as a label, and delivered with counselling. Because the test does not rely on literacy, it is less biased by education than questionnaires.

5.5.2 Assessment of Health and Safety Issues

The procedure itself has no safety risk beyond those of routine EEG. The risks are in interpretation: a false negative (about 27 % of patients) could delay care, and a false positive (about 20 % of healthy people) could cause distress and unnecessary tests. Both are controlled by using the score only to prioritise referral, by choosing the threshold for the clinical setting, and by requiring clinician

confirmation. The model was developed on one population and must be validated on Bangladeshi patients before any clinical use.

5.5.3 Assessment of Legal Issues

The data are open (CC0 licence) and de-identified, and were used and cited as the licence and the dataset authors request. Any clinical use would make the software a medical device, requiring regulatory approval and clinical validation. Patient EEG is personal health data: our fingerprinting result shows that EEG can re-identify a person across datasets, so clinical deployment must follow data-protection law, informed consent and secure storage. The open-source libraries used (MNE-Python, scikit-learn, SciPy) carry permissive licences that allow this use.

5.6 Sustainability Evaluation (PO(g))

- **Economic.** No new hardware: the method uses EEG machines already installed for epilepsy diagnosis and an ordinary computer; the software has no licence cost.
- **Environmental.** Seconds of computing per person and no consumables beyond standard EEG electrodes; far less energy and material than imaging (Section 5.2.2).
- **Technical.** Built on maintained open-source libraries, with documented, modular, seeded code that reproduces its results exactly, so it can be maintained, audited and extended by others.
- **Social.** Supports the goal of good health and well-being by making early screening more accessible, and its simple, explainable features make it easier for clinicians to trust and check.

5.7 Ethical Issues (PO(h))

- **Honest analysis.** The main result and the robustness criteria were fixed before the results were seen, every run was reported (not only the best), and the headline was tested against chance.
- **Reporting our own mistakes.** Several errors were found and corrected openly: two filter-design faults, a phase-lag measurement that was not measuring our filter, a naive alpha-peak definition, and our first, wrong explanation of the below-chance SVMs.
- **EEG as personal data.** We could match people across two "anonymous" datasets by their EEG alone, so EEG must be treated as identifiable personal data. No attempt was made to identify anyone.
- **Responsible communication.** We make no clinical claims, describe the system as screening support, credit every data source and library, and have prepared a note for the dataset authors about the participant-ID mismatch.
- **Fairness.** We checked that the model does not exploit the sex imbalance and that cleaning does not remove more data from one group than another.

6 Reflection on Individual and Team work (PO(i))

Each member owned one stage of the signal chain from design to verification and presented it in the final demonstration, while decisions that affect the validity of the results were made together.

6.1 Individual Contribution of Each Member

Table 15. *Individual contributions*

Member	ID	Main contributions
Md. Ashique Al Arafat Nayeem	2206190	Project lead; overall design of the pipeline; feature extraction; classification and validation design (nested cross-validation, checkpoints, permutation test); main result; alpha reactivity and the dataset participant-ID analysis; live demonstration tool
Arib Salehin	2206186	Robustness and confound analyses (seeds, epoch length, rejection limits, feature subsets, age and sex); error analysis; limitations, impact, sustainability and ethics assessment
H M Hamim Al Muhit	2206169	Artefact removal (re-referencing, ICA, rejection, electrode repair); the Welch estimator and its validation; comparison with the authors' derivatives; cleaning ablation
Shahriar Nafis Rabbi	2206184	Problem formulation and specifications; literature review; filter design and verification (FIR, IIR Butterworth, 50 Hz notch, zero-phase filtering)

All members took part in design reviews, in running and checking the checkpoints, in preparing the presentation, and in writing and reviewing this report.

6.2 Mode of TeamWork

The work was divided by stage of the signal chain, so that each member owned one technical part from design to verification, while the interfaces between stages were agreed together. A stage was handed on only after its checkpoint (a test against a known answer or a reference implementation) had passed and the result had been reviewed by another member. All code, figures and results were kept in one shared project folder under version control, so every result could be traced to the code that produced it. Decisions that could bias the result, such as the main analysis and the robustness criteria, were agreed and written down by the whole team before the corresponding runs. Mistakes found during reviews were recorded and corrected openly.

6.3 Diversity Statement of Team

The team combined different strengths: signal processing and filter theory, programming and data analysis, statistics and validation, and interest in the medical, legal and ethical sides of the problem; one member's parallel study of law informed the legal and ethical assessment. Different members naturally questioned different things (whether a filter met its specification, whether a statistic was fair, whether a claim could be defended), and this mix of viewpoints led directly to several of the corrections described in this report. Every member's view was heard in design decisions, and tasks were assigned by interest and ability.

6.4 Log Book of Project Implementation

Table 21 lists the milestones in order; dates are to be entered from the team's records.

Table 16. *Log book of project implementation*

Week	Milestone achieved	Individual Role	Team Role	Comments
4	Dataset selected; all recordings loaded and checked	Data pipeline and manifest	Agreed scope and main question	Checkpoint passed
4,5	First group spectra and topographic maps	Exploratory analysis	Confirmed EEG slowing in AD	Checkpoint passed
5	FIR and IIR band-pass and notch designed and verified	Filter design (Rabbi)	Specification review	Checkpoint passed
6,7	Preprocessing chain built: CAR, ICA, rejection	Artefact removal (Hamim)	Checked against authors' data	Checkpoint passed
8	Filter and analysis corrections recorded and re-run	Revision of earlier stages	Peer review of all stages	Checkpoint passed
8	Own Welch estimator validated; whole cohort processed	Spectral estimation (Hamim)	Validation review	Checkpoint passed
5,9	94 regional features extracted and ranked	Feature extraction (Nayeem)	Feature definitions agreed	Checkpoint passed
10	Alpha reactivity measured; participant-ID mismatch found	Reactivity and ID analysis (Nayeem)	Decided EEG-based pairing	Checkpoint passed
11,12	Main analysis declared; 52 classification runs; AUC 0.811	Classification and validation (Nayeem)	Main analysis fixed in advance	Checkpoint passed
12,13	Feature families, cleaning ablation, error analysis	Exploratory analyses (all)	Interpretation discussed	Checkpoint passed
13	Robustness: 5 of 5 pre-declared criteria met	Robustness and confounds (Arib)	Criteria fixed in advance	Checkpoint passed
14	Final presentation and live demonstration	Each member presented one part	Rehearsals	
14	Final report completed	Sections by stage owner	Joint review	

7 Communication to External Stakeholders (PO(j))

The project was communicated to non-specialists through the press release below, a user manual, the code repository and a demonstration video.

7.1 Executive Summary

BUET students build a low-cost brain-wave test to flag Alzheimer's disease. A team of electrical engineering students at BUET has developed software that reads a standard EEG, a harmless recording of the brain's electrical activity that most hospitals can already make, and estimates the chance that a person has Alzheimer's disease. Alzheimer's slows the brain's natural rhythms, and the software measures this slowing. Tested only on people it had never seen, it ranked patients above healthy people 81 % of the time. It cannot diagnose on its own, but it could help decide cheaply who should see a specialist, especially where brain scans are scarce.

7.2 User Manual

Requirements. A computer with Python 3.10 or newer (Windows, Linux or macOS), about 10 GB of free disk space and an internet connection for the first download.

1. Install. Create a virtual environment in the project folder and install the libraries:

```
python -m venv .venv → .venv\Scripts\Activate.ps1 → pip install mne numpy scipy scikit-learn pandas matplotlib
```

2. Get the data. Download OpenNeuro ds004504 (and ds006036 for alpha reactivity) into data\raw\.. The scripts find the files through the paths defined in eegio.py.

3. Run the pipeline. Run the scripts in numerical order from the project folder, for example python src\12_batch_preprocess.py. Each script prints a checkpoint line first; if it reports FAIL, stop and fix the problem before continuing. Outputs go to data\processed\ (tables and caches) and figures\ (one subfolder per stage). The main result is printed by 27_classify.py and the permutation test by 28_permutation_test.py.

4. Score one person (demonstration). Once, beforehand: python src\44_live_demo.py --prepare (about one minute; caches the group spectra) and python src\44_live_demo.py --suggest (lists people scored confidently in cross-validation). Then: python src\44_live_demo.py sub-041. The program prints each processing step, the probability of AD, the cross-validated score of that person and the true diagnosis, opens the figure of Fig. 26 and saves it to figures\demo\.

5. Read the output. The bar on the right is the probability of AD; above the dashed line at 0.5 the EEG looks like AD, below it healthy. The output is screening support for research and teaching only, not a medical diagnosis.

7.3 Github Link

<https://github.com/ashiquealarafatnayeem/AD-EEG-Detection>

8 Project Management and Cost Analysis (PO(k))

The project was managed in stages, each closed by a checkpoint. Because it is software-only and uses open data, its costs are those of computing and, for deployment, of the EEG recorder.

8.1 Bill of Materials

The project is software-only and used existing resources, so its development cost was essentially zero (Table 22).

Table 17. *Bill of materials for the developed prototype*

Item	Source	Cost (BDT)
EEG data (ds004504, ds006036)	OpenNeuro, CC0 licence	0
Python, NumPy, SciPy, MNE-Python, scikit-learn, pandas, matplotlib	Open source	0
Development computer	Members' existing laptops	0 (existing)
Electricity (about 0.05 kWh per full cohort run, a few dozen runs)	Estimate	< 50
Total		< 50

8.2 Calculation of Per Unit Cost of Prototype

A working screening station needs an EEG recorder and a computer. In a hospital that already owns a clinical EEG machine, the extra cost per unit is only the software (free) and the existing computer. For a stand-alone research prototype, an open-hardware 16-channel EEG board costs about USD 2,499 [28] (about BDT 3 lakh at an assumed rate of BDT 120 per USD), plus an electrode cap and a laptop. Per test, the marginal cost is the technician's time (about 20–30 minutes for set-up and a 13-minute recording) and electrode gel, comparable to a routine EEG, which costs about BDT 500–1,500 in government hospitals [2].

8.3 Calculation of Per Unit Cost of Mass-Produced Unit

Software has no per-copy production cost, so for mass deployment the cost per installation is installation, training and validation, shared over many sites. Run on existing hospital EEG machines, the cost per screening is close to that of a routine EEG and an order of magnitude below MRI or PET. A dedicated few-electrode headset (posterior electrodes alone reached AUC 0.828) could be produced far more cheaply than a 19-channel clinical system, but its cost can only be estimated reliably after a prototype has been built and validated.

8.4 Timeline of Project Implementation

Fig. 43 shows the sequence of the work. Each phase ended with a checkpoint before the next one began.

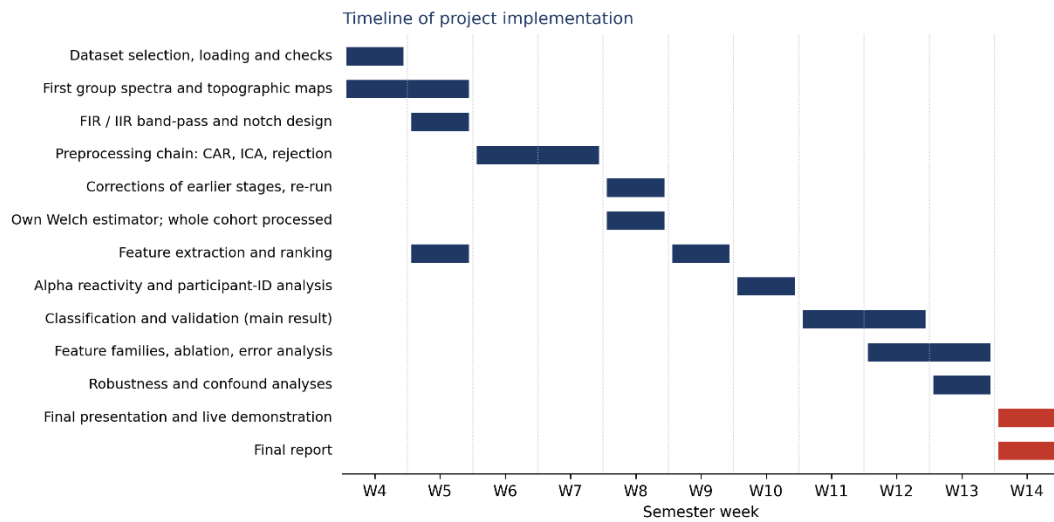


Figure 43. Gantt chart of project implementation

9 Future Work (PO(I))

The most important next steps follow directly from the limitations:

- **External validation.** Test the frozen pipeline, without retraining, on recordings from another hospital, starting with healthy older adults from a public dataset and, ideally, Bangladeshi patients.
- **Separating AD from FTD.** Add features about where changes occur: frontal against posterior contrasts, connectivity (coherence, phase-lag index) and source localisation, declared in advance and tested with the same nested protocol.
- **Earlier stages.** Apply the pipeline to mild cognitive impairment and follow the same people over time.
- **A low-cost device.** Posterior theta/alpha needs only a few electrodes, so a simple headset with embedded filtering and on-device Welch estimation is a realistic hardware project.
- **Better use of the data.** Model choice inside the inner loop, calibrated probabilities, and a decision threshold chosen for the clinical setting.

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